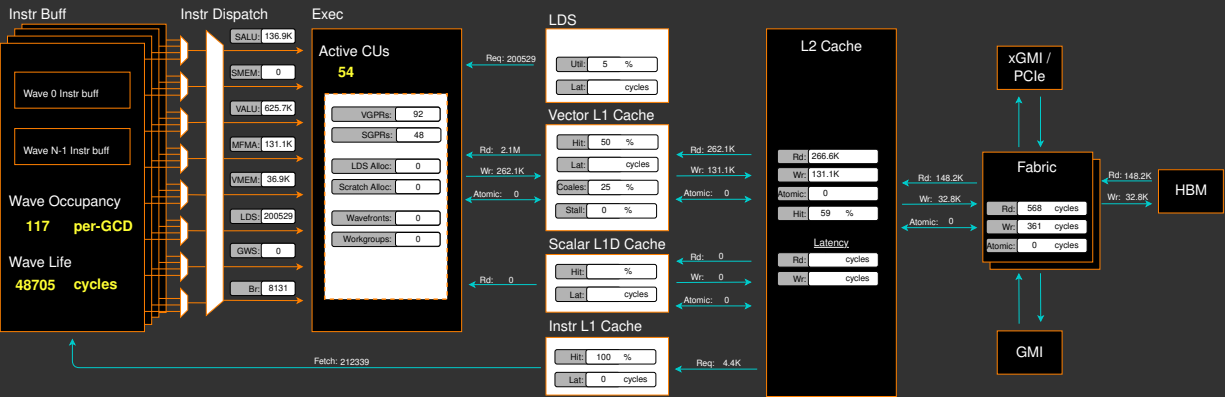
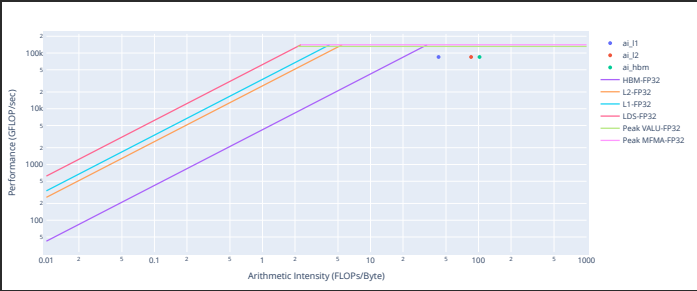


Placeholder. Guided Analysis coming soon...



Empirical Roofline Analysis (Flops)



## 10. Top Stats

## 0.1 Top Kernels

| %Kernel_Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Count   | Sum(ns)     | Mean(ns) | Median(ns) | Pct   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------------|----------|------------|-------|
| void at::native::vectorized_elementwise_kernel<4,<br>at::native::FillFunction<int*, std::array<char*, 1ul>*>int,<br>at::native::FillFunction<int*, std::array<char*, 1ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1484.00 | 59575921.13 | 42432.35 | 42342.75   | 85.66 |
| bias_kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1594.00 | 57555918.81 | 3610.99  | 3471.25    | 8.28  |
| matmul_kernel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 152.00  | 3846198.53  | 25303.94 | 24692.70   | 5.53  |
| _amd_rocclr_copyBuffer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 89.00   | 268785.00   | 3020.86  | 2986.59    | 0.39  |
| void at::native::(anonymous<br>namespace)::distribution_elementwise_grid_stride_kernel<float,<br>4,<br>at::native::templates::cuda::normal_and_transform<c10::Half,<br>float, at::CUDAGeneratorImpl*,<br>at::native::templates::cuda::normal_kernel<at::CUDAGeneratorIm<br>const&, double, double, at::CUDAGeneratorImpl*>:::<br>(lambda)#1)::operator()() const:(lambda)#3)::operator()()<br>const:::(lambda(float)#1)::(at::TensorIteratorBase&<br>at::CUDAGeneratorImpl*,<br>at::native::templates::cuda::normal_kernel<at::CUDAGeneratorIm<br>const&, double, double, at::CUDAGeneratorImpl*>:::<br>(lambda)#1)::operator()() const:(lambda)#3)::operator()()<br>const:::(lambda(float)#1)):: | 3.00    | 18722.91    | 6240.97  | 7286.66    | 0.03  |

## 0.2 Dispatch List

| Dispatch_ID | Kernel_Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | GPU_ID |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 0           | <pre> namespace:):distribution_elementwise_grid_transform_kernel&lt;float, 4, at::native::templates::cuda::normal_and_transform&lt;c10::Half, float, at::native::templates::cuda::normal_kernel_cat::CUDAGeneratorImpl*&gt;() const:TensorBase const&amp;, double, double, at::CUDAGeneratorImpl*&gt;::(lambda)#1):operator()() const:=(lambda)#3):operator()() const:(lambda(float)#1) &gt;at::TensorIteratorBase&amp;, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel_cat::CUDAGeneratorImpl*&gt;() const:TensorBase const&amp;, double, double, at::CUDAGeneratorImpl*, const:=(lambda)#3):operator()() const:(lambda(float)#1) :(lambda(hiprand_statePhilox4_32_1024) const:=(lambda(float)#1) namespace:):distribution_nullary_kernel&lt;c10::Half, float, HIP_vector_type&lt;float, 4u&gt;, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_and_transform&lt;c10::Half, float, at::native::templates::cuda::normal_kernel_cat::CUDAGeneratorImpl*&gt;() const:TensorBase const&amp;, double, double, at::CUDAGeneratorImpl*&gt;::(lambda)#1):operator()() const:=(lambda)#3):operator()() const:(lambda(float)#1) &gt;at::TensorIteratorBase&amp;, at::CUDAGeneratorImpl*, at::native::templates::cuda::normal_kernel_cat::CUDAGeneratorImpl*&gt;() const:TensorBase </pre> | 0      |

## 1. System Info

| Index                  | Info                                          |
|------------------------|-----------------------------------------------|
| workload_name          | nonfusion_matmulbias                          |
| command                | python nonfusion_matmulbias.py                |
| ip_blocks              | 5Q LD5 SQC TA TD TCP TCC SPI CPC CPF Roofline |
| timestamp              | Mon Jan 12 19:01:21 2026 (UTC)                |
| version                | 3                                             |
| hostname               | f57b369e0635                                  |
| cpu_model              | Intel(R) Xeon(R) Platinum 8470                |
| sbios                  | SeaBIOS1.15.0-1                               |
| linux_distro           | CentOS Stream 10 (Coughlan)                   |
| linux_kernel_version   | 5.15.0-164-generic                            |
| amd_gpu_kernel_version |                                               |
| cpu_memory             | 230893524                                     |
| gpu_memory             |                                               |
| rocm_version           | 7.0.3                                         |
| vbios                  | 113-M3000108-103                              |
| compute_partition      | SPX                                           |
| memory_partition       |                                               |
| gpu_series             | M1300                                         |
| gpu_model              | M1300X_A1                                     |
| gpu_arch               | gfx942                                        |
| gpu_chip_id            | 29877                                         |
| gpu_l1                 | 32                                            |
| gpu_l2                 | 4096                                          |
| cu_per_gpu             | 304                                           |
| simd_per_cu            | 4                                             |
| se_per_gpu             | 32                                            |
| wave_size              | 64                                            |
| workgroup_max_size     | 1024                                          |
| max_waves_per_cu       | 32                                            |
| max_sclk               | 2100                                          |
| max_mclk               | 1300                                          |
| cur_sclk               | 2100                                          |
| cur_mclk               | 1300                                          |
| total_l2_chan          | 128                                           |
| lds_banks_per_cu       | 32                                            |
| sqc_per_gpu            | 160                                           |
| pipes_per_gpu          | 4                                             |
| num_xcd                | 8                                             |
| num_hbm_channels       | 128                                           |

2. System Speed-of-Light

2.1 Speed-of-Light

| #Metric                   | \$ | Avg      | \$ | Unit             | \$ | Peak       | \$ | Pct of Peak |
|---------------------------|----|----------|----|------------------|----|------------|----|-------------|
| VALU FLOPs                |    | 9.97     |    | Gflop/s          |    | 81715.20   |    | 0.01        |
| VALU IOPs                 |    | 724.47   |    | Giop/s           |    | 81715.20   |    | 0.89        |
| MFMA FLOPs (F8)           |    | 0.00     |    | Gflop/s          |    | 2614886.40 |    | 0.00        |
| MFMA FLOPs (BF16)         |    | 0.00     |    | Gflop/s          |    | 1307443.20 |    | 0.00        |
| MFMA FLOPs (F16)          |    | 85978.80 |    | Gflop/s          |    | 1307443.20 |    | 6.58        |
| MFMA FLOPs (F32)          |    | 0.00     |    | Gflop/s          |    | 163430.40  |    | 0.00        |
| MFMA FLOPs (F64)          |    | 0.00     |    | Gflop/s          |    | 163430.40  |    | 0.00        |
| MFMA FLOPs (F64F4)        |    |          |    | Gflop/s          |    |            |    |             |
| MFMA IOPs (Int8)          |    | 0.00     |    | Giop/s           |    | 2614886.40 |    | 0.00        |
| Active CUs                |    | 54.00    |    | Cus              |    | 304.00     |    | 17.76       |
| SALU Utilization          |    | 0.66     |    | Pct              |    | 100.00     |    | 0.66        |
| VALU Utilization          |    | 3.02     |    | Pct              |    | 100.00     |    | 3.02        |
| MFMA Utilization          |    | 5.01     |    | Pct              |    | 100.00     |    | 5.01        |
| VMEM Utilization          |    | 0.18     |    | Pct              |    | 100.00     |    | 0.18        |
| Branch Utilization        |    | 0.11     |    | Pct              |    | 100.00     |    | 0.11        |
| VALU Active Threads       |    | 63.90    |    | Threads          |    | 64.00      |    | 99.85       |
| IPC                       |    | 0.36     |    | Instr/cycle      |    | 5.00       |    | 7.26        |
| Wavefront Occupancy       |    | 6299.35  |    | Wavefronts       |    | 9728.00    |    | 64.75       |
| Theoretical LDS Bandwidth |    | 5238.44  |    | Gb/s             |    | 81715.20   |    | 6.41        |
| LDS Bank Conflicts/Access |    | 0.00     |    | Conflicts/access |    | 32.00      |    | 0.00        |
| vL1D Cache Hit Rate       |    | 50.00    |    | Pct              |    | 100.00     |    | 50.00       |
| vL1D Cache BW             |    | 4030.26  |    | Gb/s             |    | 81715.20   |    | 4.93        |
| L2 Cache Hit Rate         |    | 58.70    |    | Pct              |    | 100.00     |    | 58.70       |
| L2 Cache BW               |    | 2037.30  |    | Gb/s             |    | 34406.40   |    | 5.92        |
| L2-Fabric Read BW         |    | 757.51   |    | Gb/s             |    | 5324.80    |    | 14.23       |
| L2-Fabric Write BW        |    | 83.96    |    | Gb/s             |    | 5324.80    |    | 1.50        |
| L2-Fabric Read Latency    |    | 567.96   |    | Cycles           |    |            |    |             |
| L2-Fabric Write Latency   |    | 361.04   |    | Cycles           |    |            |    |             |
| sL1D Cache Hit Rate       |    |          |    | Pct              |    | 100.00     |    |             |
| sL1D Cache BW             |    | 0.00     |    | Gb/s             |    | 21504.00   |    | 0.00        |
| L1I Hit Rate              |    | 99.95    |    | Pct              |    | 100.00     |    | 99.95       |
| L1I BW                    |    | 546.95   |    | Gb/s             |    | 21504.00   |    | 2.54        |
| L1I Fetch Latency         |    |          |    | Cycles           |    |            |    |             |

5. Command Processor (CPC/CPF)

5.1 Command Processor Fetcher

| #Metric            | \$ | Avg    | \$ | Min    | \$ | Max    | \$ | Unit |
|--------------------|----|--------|----|--------|----|--------|----|------|
| CPF Utilization    |    | 100.00 |    | 100.00 |    | 100.00 |    | Pct  |
| CPF Stall          |    | 1.20   |    | 0.09   |    | 2.07   |    | Pct  |
| CPF-L2 Utilization |    | 12.95  |    | 3.48   |    | 18.43  |    | Pct  |
| CPF-L2 Stall       |    | 0.00   |    | 0.00   |    | 0.00   |    | Pct  |
| CPF-UTCL1 Stall    |    | 1.04   |    | 0.09   |    | 2.18   |    | Pct  |

5.2 Packet Processor

| #Metric                           | \$ | Avg    | \$ | Min    | \$ | Max    | \$ | Unit |
|-----------------------------------|----|--------|----|--------|----|--------|----|------|
| CPC SYNC FIFO Full Rate           |    |        |    |        |    |        |    | Pct  |
| CPC CANE Stall Rate               |    |        |    |        |    |        |    | Pct  |
| CPC ADC Utilization               |    |        |    |        |    |        |    | Pct  |
| CPC Utilization                   |    | 100.00 |    | 100.00 |    | 100.00 |    | Pct  |
| CPC Stall Rate                    |    | 26.55  |    | 23.61  |    | 31.11  |    | Pct  |
| CPC Packet Decoding Utilization   |    | 92.25  |    | 82.94  |    | 96.92  |    | Pct  |
| CPC-Workgroup Manager Utilization |    | 0.74   |    | 0.55   |    | 0.86   |    | Pct  |
| CPC-L2 Utilization                |    | 0.37   |    | 0.28   |    | 0.43   |    | Pct  |
| CPC-UTCL1 Stall                   |    | 0.24   |    | 0.17   |    | 1.72   |    | Pct  |
| CPC-UTCL2 Utilization             |    | 0.23   |    | 0.16   |    | 1.67   |    | Pct  |

6. Workgroup Manager (SPI)

6.1 Workgroup Manager Utilizations

| #Metric                         | \$ | Avg    | \$ | Min    | \$ | Max    | \$ | Unit        |
|---------------------------------|----|--------|----|--------|----|--------|----|-------------|
| Schedule-Pipe Wave Occupancy    |    |        |    |        |    |        |    | Wave        |
| Accelerator Utilization         |    | 100.00 |    | 100.00 |    | 100.00 |    | Pct         |
| Scheduler-Pipe Utilization      |    | 0.00   |    | 0.00   |    | 0.00   |    | Pct         |
| Scheduler-Pipe Wave Utilization |    |        |    |        |    |        |    | Pct         |
| Workgroup Manager Utilization   |    | 74.82  |    | 69.39  |    | 80.49  |    | Pct         |
| Shader Engine Utilization       |    | 70.35  |    | 65.14  |    | 81.45  |    | Pct         |
| SIMD Utilization                |    | 15.16  |    | 12.52  |    | 87.71  |    | Pct         |
| Dispatched Workgroups           |    | 0.00   |    | 0.00   |    | 0.00   |    | Workgroups  |
| Dispatched Wavefronts           |    | 0.00   |    | 0.00   |    | 0.00   |    | Wavefronts  |
| VGPR Writes                     |    |        |    |        |    |        |    | Cycles/wave |
| SGPR Writes                     |    |        |    |        |    |        |    | Cycles/wave |

6.2 Workgroup Manager - Resource Allocation

| #Metric                                | \$ | Avg  | \$ | Min  | \$ | Max  | \$ | Unit |
|----------------------------------------|----|------|----|------|----|------|----|------|
| Not-scheduled Rate (Workgroup Manager) |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Not-scheduled Rate (Scheduler-Pipe)    |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Scheduler-Pipe FIFO Full Rate          |    |      |    |      |    |      |    | Pct  |
| Scheduler-Pipe Stall Rate              |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Scratch Stall Rate                     |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Insufficient SIMD Waveslots            |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Insufficient SIMD VGPRs                |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Insufficient SIMD SGPRs                |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Insufficient CU LDS                    |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Insufficient CU Barriers               |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Reached CU Workgroup Limit             |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |
| Reached CU Wavefront Limit             |    | 0.00 |    | 0.00 |    | 0.00 |    | Pct  |

7. Wavefront

7.1 Wavefront Launch Stats

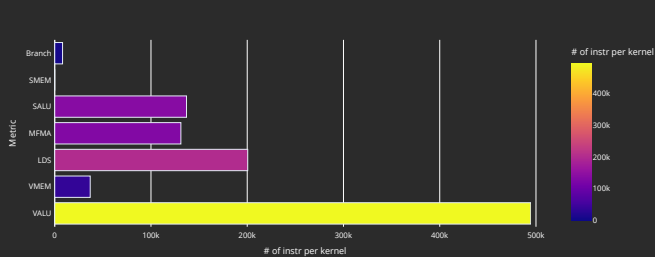
| #Metric             | ⚡        | Avg⚡     | Min⚡     | Max⚡ | Unit           |
|---------------------|----------|----------|----------|------|----------------|
| Grid Size           | 38612.21 | 16384.00 | 32768.00 |      | Work items     |
| Workgroup Size      | 478.32   | 256.00   | 512.00   |      | Work items     |
| Total Wavefronts    | 0.00     | 0.00     | 0.00     |      | Wavefronts     |
| Saved Wavefronts    | 0.00     | 0.00     | 0.00     |      | Wavefronts     |
| Restored Wavefronts | 0.00     | 0.00     | 0.00     |      | Wavefronts     |
| VGPRs               | 92.42    | 16.00    | 104.00   |      | Registers      |
| AGPRs               | 25.26    | 0.00     | 192.00   |      | Registers      |
| SGPRs               | 48.00    | 48.00    | 48.00    |      | Registers      |
| LDS Allocation      | 0.00     | 0.00     | 0.00     |      | Bytes          |
| Scratch Allocation  | 0.00     | 0.00     | 0.00     |      | Bytes/workitem |

7.2 Wavefront Runtime Stats

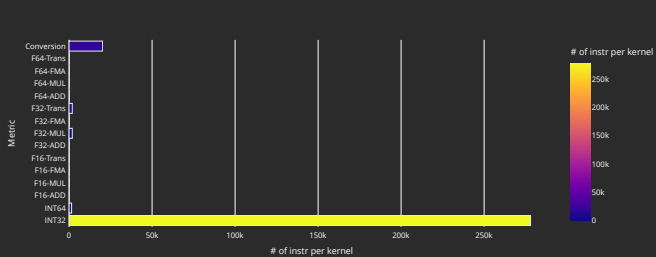
| #Metric                    | ⚡           | Avg⚡        | Min⚡        | Max⚡ | Unit              |
|----------------------------|-------------|-------------|-------------|------|-------------------|
| Kernel Time (Nanosec)      | 25383.94    | 22320.34    | 42895.66    |      | Ns                |
| Kernel Time (Cycles)       | 69356.09    | 59977.00    | 89067.63    |      | Cycle             |
| Instructions per wavefront | 2395.66     | 2220.00     | 3555.00     |      | Instr/wavefront   |
| Wave Cycles                | 22391855.45 | 14899084.00 | 84280748.00 |      | Cycles per kernel |
| Dependency Wait Cycles     | 11198316.71 | 3316048.00  | 13484544.00 |      | Cycles per kernel |
| Issue Wait Cycles          | 5899613.45  | 580628.00   | 8217988.00  |      | Cycles per kernel |
| Active Cycles              | 4644498.50  | 3849180.00  | 4766176.00  |      | Cycles per kernel |
| Wavefront Occupancy        | 6299.35     | 170.27      | 11547.34    |      | Wavefronts        |

10. Compute Units - Instruction Mix

10.1 Overall Instruction Mix



10.2 VALU Arithmetic Instr Mix



10.3 VMEM Instr Mix

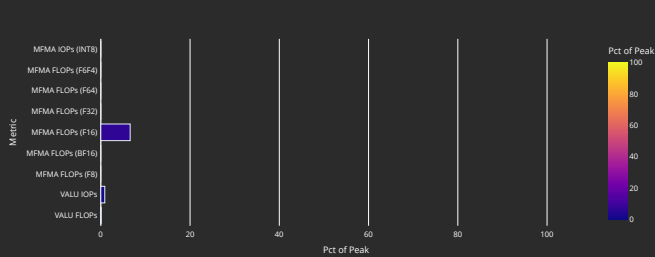
| #Metric                        | ⚡        | Avg⚡     | Min⚡     | Max⚡ | Unit             |
|--------------------------------|----------|----------|----------|------|------------------|
| Global/Generic Instr           | 0.00     | 0.00     | 0.00     |      | Instr per kernel |
| Global/Generic Read            | 0.00     | 0.00     | 0.00     |      | Instr per kernel |
| Global/Generic Write           | 0.00     | 0.00     | 0.00     |      | Instr per kernel |
| Global/Generic Atomic          | 0.00     | 0.00     | 0.00     |      | Instr per kernel |
| Spill/Stack Instr              | 36864.00 | 36864.00 | 36864.00 |      | Instr per kernel |
| Spill/Stack Coalesceable Instr |          |          |          |      | Instr per kernel |
| Spill/Stack Read               | 32768.00 | 32768.00 | 32768.00 |      | Instr per kernel |
| Spill/Stack Write              | 4096.00  | 4096.00  | 4096.00  |      | Instr per kernel |
| Spill/Stack Atomic             | 0.00     | 0.00     | 0.00     |      | Instr per kernel |

10.4 MFMA Arithmetic Instr Mix

| #Metric   | ⚡         | Avg⚡      | Min⚡      | Max⚡ | Unit             |
|-----------|-----------|-----------|-----------|------|------------------|
| MFMA-I8   | 0.00      | 0.00      | 0.00      |      | Instr per kernel |
| MFMA-F8   | 0.00      | 0.00      | 0.00      |      | Instr per kernel |
| MFMA-F16  | 131072.00 | 131072.00 | 131072.00 |      | Instr per kernel |
| MFMA-BF16 | 0.00      | 0.00      | 0.00      |      | Instr per kernel |
| MFMA-F32  | 0.00      | 0.00      | 0.00      |      | Instr per kernel |
| MFMA-F64  | 0.00      | 0.00      | 0.00      |      | Instr per kernel |
| MFMA-FGF4 |           |           |           |      | Instr per kernel |

11. Compute Units - Compute Pipeline

11.1 Speed-of-Light



11.2 Pipeline Stats

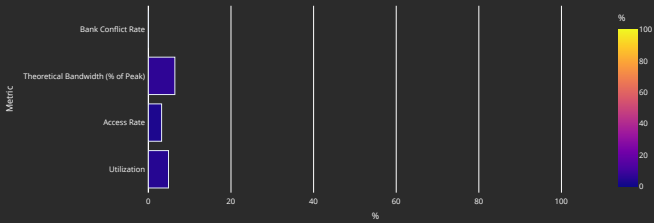
| #Metric                  | ⚡     | Avg⚡  | Min⚡  | Max⚡ | Unit         |
|--------------------------|-------|-------|-------|------|--------------|
| IPC                      | 0.36  | 0.04  | 0.43  |      | Instr/cycle  |
| IPC (Issued)             | 0.87  | 0.85  | 0.87  |      | Instr/cycle  |
| SALU Utilization         | 0.66  | 0.29  | 0.88  |      | Pct          |
| VALU Utilization         | 3.82  | 2.16  | 3.50  |      | Pct          |
| VALU Co-Issue Efficiency |       |       |       |      | Pct          |
| VMEM Utilization         | 0.18  | 0.14  | 0.20  |      | Pct          |
| Branch Utilization       | 0.11  | 0.05  | 0.13  |      | Pct          |
| VALU Active Threads      | 63.90 | 63.90 | 63.94 |      | Threads      |
| MFMA Utilization         | 5.81  | 3.87  | 5.75  |      | Pct          |
| MFMA Instr Cycles        | 32.00 | 32.00 | 32.00 |      | Cycles/instr |
| VMEM Latency             |       |       |       |      | Cycles       |
| SMEM Latency             |       |       |       |      | Cycles       |

11.3 Arithmetic Operations

| #Metric       | ⚡             | Avg⚡          | Min⚡          | Max⚡ | Unit           |
|---------------|---------------|---------------|---------------|------|----------------|
| FLOPs (Total) | 2147728545.68 | 2147614720.00 | 2147745792.00 |      | Ops per kernel |
| IOPs (Total)  | 17885722.95   | 12009472.00   | 18776864.00   |      | Ops per kernel |
| F8 OPs        | 0.00          | 0.00          | 0.00          |      | Ops per kernel |
| F16 OPs       | 2147483648.00 | 2147483648.00 | 2147483648.00 |      | Ops per kernel |
| BF16 OPs      | 0.00          | 0.00          | 0.00          |      | Ops per kernel |
| F32 OPs       | 244897.68     | 131072.00     | 262144.00     |      | Ops per kernel |
| F64 OPs       | 0.00          | 0.00          | 0.00          |      | Ops per kernel |
| FGF4 OPs      |               |               |               |      | Ops per kernel |
| INT8 OPs      | 0.00          | 0.00          | 0.00          |      | Ops per kernel |

12. Local Data Share (LDS)

12.1 Speed-of-Light

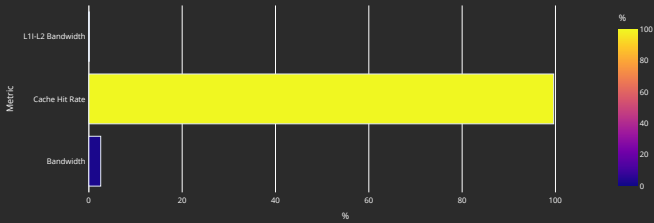


12.2 LDS Stats

| Metric                     | Avg          | Min          | Max          | Unit                |
|----------------------------|--------------|--------------|--------------|---------------------|
| LDS Instrs                 | 288528.84    | 118272.00    | 212992.00    | Instr per kernel    |
| LDS LOAD                   |              |              |              | Instr per kernel    |
| LDS STORE                  |              |              |              | Instr per kernel    |
| LDS ATOMIC                 |              |              |              | Instr per kernel    |
| LDS LOAD Bandwidth         |              |              |              | Gbps                |
| LDS STORE Bandwidth        |              |              |              | Gbps                |
| LDS ATOMIC Bandwidth       |              |              |              | Gbps                |
| Theoretical Bandwidth      | 129882671.16 | 108663296.00 | 134217728.00 | Bytes per kernel    |
| LDS Latency                |              |              |              | Cycles              |
| Bank Conflicts/Access      | 0.00         | 0.00         | 0.00         | Conflicts/access    |
| Index Accesses             | 1014083.37   | 786432.00    | 1848576.00   | Cycles per kernel   |
| Atomic Return Cycles       | 0.00         | 0.00         | 0.00         | Cycles per kernel   |
| Bank Conflict              | 0.00         | 0.00         | 0.00         | Cycles per kernel   |
| Addr Conflict              | 0.00         | 0.00         | 0.00         | Cycles per kernel   |
| Unaligned Stall            | 0.00         | 0.00         | 0.00         | Cycles per kernel   |
| Mem Violations             | 0.00         | 0.00         | 0.00         | Accesses per kernel |
| LDS Command FIFO Full Rate |              |              |              | Cycles per kernel   |
| LDS Data FIFO Full Rate    |              |              |              | Cycles per kernel   |

13. Instruction Cache

13.1 Speed-of-Light



13.2 Instruction Cache Accesses

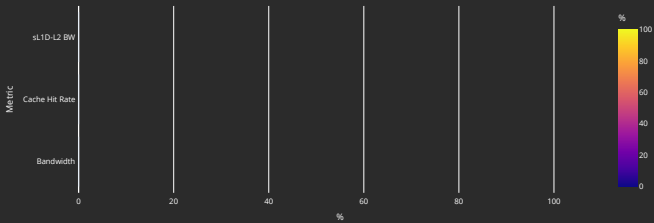
| Metric                    | Avg       | Min       | Max       | Unit              |
|---------------------------|-----------|-----------|-----------|-------------------|
| Req                       | 212338.53 | 180992.00 | 217888.00 | Req per kernel    |
| Hits                      | 211521.09 | 180159.00 | 216341.00 | Hits per kernel   |
| Misses - Non Duplicated   | 99.31     | 80.00     | 168.00    | Misses per kernel |
| Misses - Duplicated       | 714.57    | 604.00    | 863.00    | Misses per kernel |
| Cache Hit Rate            | 99.62     | 99.56     | 99.66     | Pct               |
| Instruction Fetch Latency |           |           |           | Cycles            |

13.3 Instruction Cache - L2 Interface

| Metric           | Avg       | Min       | Max       | Unit             |
|------------------|-----------|-----------|-----------|------------------|
| L1I-L2 Bandwidth | 283809.68 | 209920.00 | 515584.00 | Bytes per kernel |

14. Scalar L1 Data Cache

14.1 Speed-of-Light



14.2 Scalar L1D Cache Accesses

| Metric                  | Avg  | Min  | Max  | Unit           |
|-------------------------|------|------|------|----------------|
| Req                     | 0.00 | 0.00 | 0.00 | Req per kernel |
| Hits                    | 0.00 | 0.00 | 0.00 | Req per kernel |
| Misses - Non Duplicated | 0.00 | 0.00 | 0.00 | Req per kernel |
| Misses- Duplicated      | 0.00 | 0.00 | 0.00 | Req per kernel |
| Cache Hit Rate          |      |      |      | Pct            |
| Read Req (Total)        | 0.00 | 0.00 | 0.00 | Req per kernel |
| Atomic Req              | 0.00 | 0.00 | 0.00 | Req per kernel |
| Read Req (1 DWord)      | 0.00 | 0.00 | 0.00 | Req per kernel |
| Read Req (2 DWord)      | 0.00 | 0.00 | 0.00 | Req per kernel |
| Read Req (4 DWord)      | 0.00 | 0.00 | 0.00 | Req per kernel |
| Read Req (8 DWord)      | 0.00 | 0.00 | 0.00 | Req per kernel |
| Read Req (16 DWord)     | 0.00 | 0.00 | 0.00 | Req per kernel |

14.3 Scalar L1D Cache - L2 Interface

| Metric       | Avg  | Min  | Max  | Unit              |
|--------------|------|------|------|-------------------|
| sL1D-L2 BW   | 0.00 | 0.00 | 0.00 | Bytes per kernel  |
| Read Req     | 0.00 | 0.00 | 0.00 | Req per kernel    |
| Write Req    | 0.00 | 0.00 | 0.00 | Req per kernel    |
| Atomic Req   | 0.00 | 0.00 | 0.00 | Req per kernel    |
| Stall Cycles | 0.00 | 0.00 | 0.00 | Cycles per kernel |

15. Address Processing Unit and Data Return Path (TA/TD)

15.1 Address Processing Unit

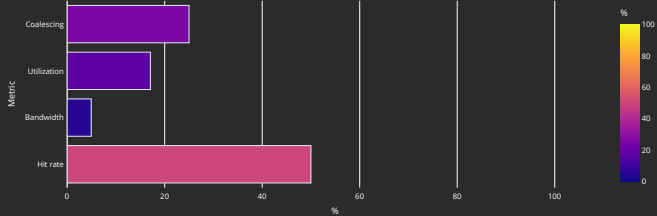
| Metric                                   | Avg       | Min       | Max       | Unit                    |
|------------------------------------------|-----------|-----------|-----------|-------------------------|
| Address Processing Unit Busy             | 8.20      | 6.59      | 9.22      | Pct                     |
| Address Stall                            | 5.31      | 2.88      | 6.64      | Pct                     |
| Data Stall                               | 2.14      | 1.83      | 2.93      | Pct                     |
| Data-Processor - Address Stall           | 0.00      | 0.00      | 0.00      | Pct                     |
| Sequencer - TA Address Stall             |           |           |           | Cycles per kernel       |
| Sequencer - TA Command Stall             |           |           |           | Cycles per kernel       |
| Sequencer - TA Data Stall                |           |           |           | Cycles per kernel       |
| Total Instructions                       | 36864.00  | 36864.00  | 36864.00  | Instructions per kernel |
| Global/Generic Instructions              | 0.00      | 0.00      | 0.00      | Instructions per kernel |
| Global/Generic Read Instructions         | 0.00      | 0.00      | 0.00      | Instructions per kernel |
| Global/Generic Read Instructions for LDS |           |           |           | Instructions per kernel |
| Global/Generic Write Instructions        | 0.00      | 0.00      | 0.00      | Instructions per kernel |
| Global/Generic Atomic Instructions       | 0.00      | 0.00      | 0.00      | Instructions per kernel |
| Spill/Stack Instructions                 | 36864.00  | 36864.00  | 36864.00  | Instructions per kernel |
| Spill/Stack Read Instructions            | 32768.00  | 32768.00  | 32768.00  | Instructions per kernel |
| Spill/Stack Read Instructions for LDS    |           |           |           | Instructions per kernel |
| Spill/Stack Write Instructions           | 4096.00   | 4096.00   | 4096.00   | Instructions per kernel |
| Spill/Stack Atomic Instructions          | 0.00      | 0.00      | 0.00      | Instructions per kernel |
| Spill/Stack Total Cycles                 | 589824.00 | 589824.00 | 589824.00 | Cycles per kernel       |
| Spill/Stack Coalesced Read               | 0.00      | 0.00      | 0.00      | Cycles per kernel       |
| Spill/Stack Coalesced Write              | 0.00      | 0.00      | 0.00      | Cycles per kernel       |

15.2 Data-Return Path

| Metric                                | Avg      | Min      | Max      | Unit                    |
|---------------------------------------|----------|----------|----------|-------------------------|
| Data-Return Busy                      | 12.91    | 9.31     | 14.22    | Pct                     |
| Cache RAM - Data-Return Stall         | 10.28    | 7.21     | 11.96    | Pct                     |
| Workgroup manager - Data-Return Stall | 0.00     | 0.00     | 0.00     | Pct                     |
| Coalescable Instructions              | 0.00     | 0.00     | 0.00     | Instructions per kernel |
| Read Instructions                     | 32768.00 | 32768.00 | 32768.00 | Instructions per kernel |
| Write Instructions                    | 4096.00  | 4096.00  | 4096.00  | Instructions per kernel |
| Atomic Instructions                   | 0.00     | 0.00     | 0.00     | Instructions per kernel |
| Write Ack Instructions                |          |          |          | Instructions per kernel |

16. Vector L1 Data Cache

16.1 Speed-of-Light



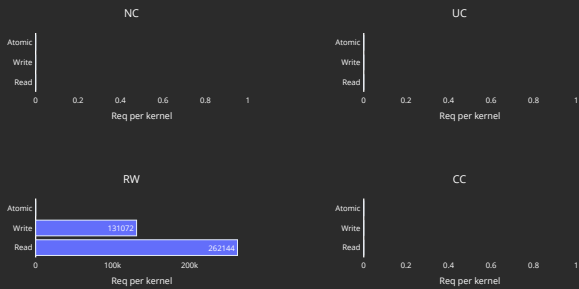
16.2 L1D Cache Stalls (%)

| Metric                  | Min  | Q1    | Median | Q3    | Max   |
|-------------------------|------|-------|--------|-------|-------|
| Stalled on L2 Data      | 9.83 | 12.51 | 13.36  | 13.54 | 14.44 |
| Stalled on L2 Req       | 0.00 | 0.00  | 0.00   | 0.00  | 0.00  |
| Stalled on Address      |      |       |        |       |       |
| Stalled on Data         |      |       |        |       |       |
| Stalled on Latency FIFO |      |       |        |       |       |
| Stalled on Request FIFO |      |       |        |       |       |
| Stalled on Read Return  |      |       |        |       |       |
| Tag RAM Stall (Read)    | 0.00 | 0.00  | 0.00   | 0.00  | 0.00  |
| Tag RAM Stall (Write)   | 0.92 | 1.21  | 1.22   | 1.27  | 1.40  |
| Tag RAM Stall (Atomic)  | 0.00 | 0.00  | 0.00   | 0.00  | 0.00  |

16.3 L1D Cache Accesses

| Metric              | Avg          | Min          | Max          | Unit             |
|---------------------|--------------|--------------|--------------|------------------|
| Total Req           | 2359296.00   | 2359296.00   | 2359296.00   | Req per kernel   |
| Read Req            | 2097152.00   | 2097152.00   | 2097152.00   | Req per kernel   |
| Write Req           | 262144.00    | 262144.00    | 262144.00    | Req per kernel   |
| Atomic Req          | 0.00         | 0.00         | 0.00         | Req per kernel   |
| Cache BW            | 100663296.00 | 100663296.00 | 100663296.00 | Bytes per kernel |
| Cache Hit Rate      | 50.00        | 50.00        | 50.00        | Pct              |
| Cache Accesses      | 786432.00    | 786432.00    | 786432.00    | Req per kernel   |
| Cache Hits          | 393216.00    | 393216.00    | 393216.00    | Req per kernel   |
| Invalidations       | 64.00        | 64.00        | 64.00        | Req per kernel   |
| L1-L2 BW            | 41943040.00  | 41943040.00  | 41943040.00  | Bytes per kernel |
| Tag RAM 0 Req       |              |              |              | Req per kernel   |
| Tag RAM 1 Req       |              |              |              | Req per kernel   |
| Tag RAM 2 Req       |              |              |              | Req per kernel   |
| Tag RAM 3 Req       |              |              |              | Req per kernel   |
| L1-L2 Read          | 262144.00    | 262144.00    | 262144.00    | Req per kernel   |
| L1-L2 Write         | 131072.00    | 131072.00    | 131072.00    | Req per kernel   |
| L1-L2 Atomic        | 0.00         | 0.00         | 0.00         | Req per kernel   |
| L1 Access Latency   |              |              |              | Cycles           |
| L1-L2 Read Latency  |              |              |              | Cycles           |
| L1-L2 Write Latency |              |              |              | Cycles           |

16.4 L1D - L2 Transactions



16.5 L1D Addr Translation

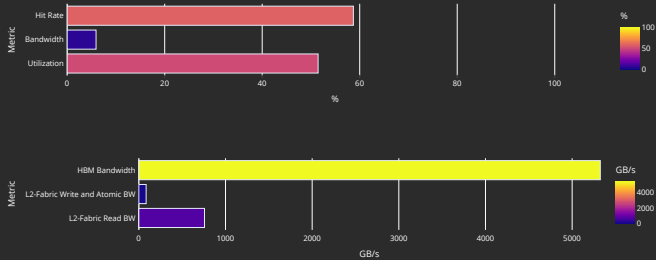
| Metric                        | Avg       | Min       | Max       | Units          |
|-------------------------------|-----------|-----------|-----------|----------------|
| Req                           | 786432.00 | 786432.00 | 786432.00 | Req per kernel |
| Inflight Req                  |           |           |           | Req per kernel |
| Hit Ratio                     | 99.97     | 98.98     | 100.00    | Pct            |
| Hits                          | 786176.41 | 778430.00 | 786432.00 | Req per kernel |
| Translation Misses            | 151.37    | 0.00      | 192.00    | Req per kernel |
| Misses under Translation Miss |           |           |           | Req per kernel |
| Permission Misses             | 0.84      | 0.00      | 128.00    | Req per kernel |

16.6 L1D Addr Translation Stalls

| Metric                   | Avg | Min | Max | Units             |
|--------------------------|-----|-----|-----|-------------------|
| Cache Full Stall         |     |     |     | Cycles per kernel |
| Cache Miss Stall         |     |     |     | Cycles per kernel |
| Serialization Stall      |     |     |     | Cycles per kernel |
| Thrashing Stall          |     |     |     | Cycles per kernel |
| Latency FIFO Stall       |     |     |     | Cycles per kernel |
| Resident Page Full Stall |     |     |     | Cycles per kernel |
| UTCL2 Stall              |     |     |     | Cycles per kernel |

17. L2 Cache

17.1 Speed-of-Light



17.2 L2 - Fabric Transactions

| Metric                            | Avg         | Min         | Max         | Unit             |
|-----------------------------------|-------------|-------------|-------------|------------------|
| Read BW                           | 18921411.37 | 18916352.00 | 18954240.00 | Bytes per kernel |
| HBM Read Traffic                  | 100.00      | 99.98       | 100.01      | Pct              |
| Remote Read Traffic               | 0.00        | 0.00        | 0.02        | Pct              |
| Uncached Read Traffic             | 0.01        | 0.00        | 0.02        | Pct              |
| Write and Atomic BW               | 2097152.00  | 2097152.00  | 2097152.00  | Bytes per kernel |
| HBM Write and Atomic Traffic      | 100.00      | 100.00      | 100.00      | Pct              |
| Remote Write and Atomic Traffic   | 0.00        | 0.00        | 0.00        | Pct              |
| Atomic Traffic                    | 0.00        | 0.00        | 0.00        | Pct              |
| Uncached Write and Atomic Traffic | 0.00        | 0.00        | 0.00        | Pct              |
| Read Latency                      | 567.96      | 509.34      | 1144.28     | Cycles           |
| Write and Atomic Latency          | 361.04      | 317.26      | 1058.34     | Cycles           |
| Atomic Latency                    |             |             |             | Cycles           |
| Read Stall                        |             |             |             | Pct              |
| Write Stall                       |             |             |             | Pct              |

17.3 L2 Cache Accesses

| Metric               | Avg         | Min         | Max         | Unit                  |
|----------------------|-------------|-------------|-------------|-----------------------|
| Bandwidth            | 50899092.21 | 50772480.00 | 51364864.00 | Bytes per kernel      |
| Read Bandwidth       |             |             |             | Bytes per kernel      |
| Write Bandwidth      |             |             |             | Bytes per kernel      |
| Atomic Bandwidth     |             |             |             | Bytes per kernel      |
| Req                  | 397649.16   | 396660.00   | 401288.00   | Req per kernel        |
| Read Req             | 266598.07   | 265506.00   | 270048.00   | Req per kernel        |
| Write Req            | 131072.00   | 131072.00   | 131072.00   | Req per kernel        |
| Atomic Req           | 0.00        | 0.00        | 0.00        | Req per kernel        |
| Streaming Req        | 0.00        | 0.00        | 0.00        | Req per kernel        |
| Bypasss Req          |             |             |             | Req per kernel        |
| Probe Req            | 0.00        | 0.00        | 0.00        | Req per kernel        |
| Input Buffer Req     |             |             |             | Req per kernel        |
| Cache Hit            | 58.70       | 58.61       | 59.02       | Pct                   |
| Hits                 | 233389.80   | 232492.00   | 236824.00   | Hits per kernel       |
| Misses               | 164213.80   | 164168.00   | 164464.00   | Misses per kernel     |
| Writeback            | 16496.53    | 16496.00    | 16504.00    | Cachelines per kernel |
| Writeback (Internal) | 0.00        | 0.00        | 0.00        | Cachelines per kernel |
| Writeback (vLID Req) | 16384.00    | 16384.00    | 16384.00    | Cachelines per kernel |
| Evict (Internal)     | 0.00        | 0.00        | 0.00        | Cachelines per kernel |
| Evict (vLID Req)     | 0.00        | 0.00        | 0.00        | Cachelines per kernel |
| NC Req               | 0.00        | 0.00        | 0.00        | Req per kernel        |
| UC Req               | 5.69        | 0.00        | 16.00       | Req per kernel        |
| CC Req               | 0.00        | 0.00        | 0.00        | Req per kernel        |
| RW Req               | 397473.32   | 396496.00   | 401728.00   | Req per kernel        |

17.4 L2 Cache Stalls

| Metric                     | Avg | Min | Max | Unit              |
|----------------------------|-----|-----|-----|-------------------|
| Stalled on Latency FIFO    |     |     |     | Cycles per kernel |
| Stalled on Write Data FIFO |     |     |     | Cycles per kernel |
| Input Buffer Stalled on L2 |     |     |     | Cycles per kernel |

17.5 L2 - Fabric Interface Stalls



17.6 L2 - Fabric Detailed Transaction Breakdown

| Metric                              | Avg       | Min       | Max       | Unit             |
|-------------------------------------|-----------|-----------|-----------|------------------|
| Read (32B)                          | 0.00      | 0.00      | 0.00      | Req per kernel   |
| Read (64B)                          | 735.05    | 656.00    | 1248.00   | Req per kernel   |
| Read (128B)                         | 147456.00 | 147456.00 | 147456.00 | Req per kernel   |
| Read (Uncached)                     | 11.38     | 0.00      | 32.00     | Req per kernel   |
| HBM Read                            | 148188.11 | 148112.00 | 148688.00 | Req per kernel   |
| Remote Read                         | 3.68      | 0.00      | 32.00     | Req per kernel   |
| Read Bandwidth - PCIe               |           |           |           | Bytes per kernel |
| Read Bandwidth - Infinity Fabric™   |           |           |           | Bytes per kernel |
| Read Bandwidth - HBM                |           |           |           | Bytes per kernel |
| Write and Atomic (32B)              | 0.00      | 0.00      | 0.00      | Req per kernel   |
| Write and Atomic (Uncached)         | 0.00      | 0.00      | 0.00      | Req per kernel   |
| Write and Atomic (64B)              | 32768.00  | 32768.00  | 32768.00  | Req per kernel   |
| HBM Write and Atomic                | 32768.00  | 32768.00  | 32768.00  | Req per kernel   |
| Remote Write and Atomic             | 0.00      | 0.00      | 0.00      | Req per kernel   |
| Write Bandwidth - PCIe              |           |           |           | Bytes per kernel |
| Write Bandwidth - Infinity Fabric™  |           |           |           | Bytes per kernel |
| Write Bandwidth - HBM               |           |           |           | Bytes per kernel |
| Atomic                              | 0.00      | 0.00      | 0.00      | Req per kernel   |
| Atomic - HBM                        |           |           |           | Req per kernel   |
| Atomic Bandwidth - PCIe             |           |           |           | Bytes per kernel |
| Atomic Bandwidth - Infinity Fabric™ |           |           |           | Bytes per kernel |
| Atomic Bandwidth - HBM              |           |           |           | Bytes per kernel |



18. L2 Cache (per Channel)

18.1 Aggregate Stats (All channels)

| Metric            | Avg   | Std Dev | Min   | Max   | Unit |
|-------------------|-------|---------|-------|-------|------|
| L2 Cache Hit Rate | 58.78 | 0.13    | 58.52 | 59.19 | Pct  |

18.2 L2 Cache Hit Rate (pct)

| Channel | Min   | Q1    | Median | Q3    | Max   |
|---------|-------|-------|--------|-------|-------|
| 0.00    | 58.45 | 58.50 | 58.55  | 58.61 | 58.66 |
| 1.00    | 58.43 | 58.50 | 58.55  | 58.61 | 58.66 |
| 2.00    | 58.45 | 58.50 | 58.55  | 58.61 | 59.02 |
| 3.00    | 58.54 | 58.62 | 58.70  | 58.78 | 59.02 |
| 4.00    | 58.65 | 58.78 | 58.87  | 58.97 | 59.09 |
| 5.00    | 58.67 | 58.78 | 58.88  | 58.97 | 59.09 |
| 6.00    | 58.65 | 58.76 | 58.87  | 58.95 | 59.07 |
| 7.00    | 58.65 | 58.78 | 58.88  | 58.99 | 59.09 |
| 8.00    | 58.47 | 58.54 | 58.59  | 58.65 | 59.44 |
| 9.00    | 58.48 | 58.54 | 58.59  | 58.65 | 59.44 |
| 10.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.44 |
| 11.00   | 58.47 | 58.54 | 58.59  | 58.65 | 59.44 |
| 12.00   | 58.47 | 58.54 | 58.59  | 58.65 | 59.44 |
| 13.00   | 58.47 | 58.54 | 58.59  | 58.65 | 59.44 |
| 14.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.48 |
| 15.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.48 |
| 16.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.44 |
| 17.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.44 |
| 18.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.44 |
| 19.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.44 |
| 20.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.44 |
| 21.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.44 |
| 22.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.48 |
| 23.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.48 |
| 24.00   | 58.41 | 58.50 | 58.55  | 58.61 | 58.66 |
| 25.00   | 58.41 | 58.50 | 58.55  | 58.61 | 58.66 |
| 26.00   | 58.45 | 58.50 | 58.55  | 58.61 | 59.02 |
| 27.00   | 58.54 | 58.62 | 58.70  | 58.78 | 59.02 |
| 28.00   | 58.60 | 58.78 | 58.88  | 58.97 | 59.09 |
| 29.00   | 58.60 | 58.78 | 58.88  | 58.99 | 59.09 |
| 30.00   | 58.60 | 58.76 | 58.87  | 58.97 | 59.07 |
| 31.00   | 58.60 | 58.78 | 58.88  | 58.99 | 59.09 |
| 32.00   | 58.45 | 58.50 | 58.55  | 58.61 | 58.66 |
| 33.00   | 58.45 | 58.50 | 58.55  | 58.61 | 58.66 |
| 34.00   | 58.45 | 58.54 | 58.55  | 58.61 | 59.02 |
| 35.00   | 58.52 | 58.68 | 58.78  | 58.78 | 59.02 |
| 36.00   | 58.65 | 58.78 | 58.88  | 58.97 | 59.09 |
| 37.00   | 58.65 | 58.78 | 58.88  | 58.99 | 59.09 |
| 38.00   | 58.63 | 58.76 | 58.87  | 58.97 | 59.07 |
| 39.00   | 58.67 | 58.78 | 58.88  | 58.99 | 59.09 |
| 40.00   | 58.47 | 58.57 | 58.59  | 58.65 | 59.44 |
| 41.00   | 58.48 | 58.59 | 58.59  | 58.65 | 59.44 |
| 42.00   | 58.48 | 58.57 | 58.59  | 58.65 | 59.44 |
| 43.00   | 58.47 | 58.57 | 58.59  | 58.65 | 59.44 |
| 44.00   | 58.48 | 58.57 | 58.59  | 58.65 | 59.44 |
| 45.00   | 58.48 | 58.57 | 58.59  | 58.65 | 59.44 |
| 46.00   | 58.47 | 58.57 | 58.59  | 58.65 | 59.48 |
| 47.00   | 58.48 | 58.57 | 58.59  | 58.65 | 59.48 |
| 48.00   | 58.47 | 58.54 | 58.59  | 58.65 | 59.29 |
| 49.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.29 |
| 50.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.29 |
| 51.00   | 58.47 | 58.54 | 58.59  | 58.65 | 59.29 |
| 52.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.29 |
| 53.00   | 58.47 | 58.54 | 58.59  | 58.65 | 59.29 |
| 54.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.33 |
| 55.00   | 58.48 | 58.54 | 58.59  | 58.65 | 59.33 |
| 56.00   | 58.41 | 58.50 | 58.55  | 58.61 | 58.66 |
| 57.00   | 58.39 | 58.50 | 58.55  | 58.61 | 58.66 |
| 58.00   | 58.45 | 58.50 | 58.55  | 58.61 | 58.91 |
| 59.00   | 58.52 | 58.62 | 58.70  | 58.78 | 58.91 |
| 60.00   | 58.60 | 58.78 | 58.87  | 58.97 | 59.09 |
| 61.00   | 58.60 | 58.78 | 58.88  | 58.99 | 59.09 |
| 62.00   | 58.60 | 58.76 | 58.87  | 58.97 | 59.07 |
| 63.00   | 58.60 | 58.78 | 58.88  | 58.99 | 59.09 |
| 64.00   | 58.47 | 58.59 | 58.65  | 58.65 | 59.44 |
| 65.00   | 58.48 | 58.59 | 58.65  | 58.65 | 59.44 |
| 66.00   | 58.48 | 58.59 | 58.63  | 58.65 | 59.44 |
| 67.00   | 58.47 | 58.57 | 58.63  | 58.65 | 59.44 |
| 68.00   | 58.48 | 58.59 | 58.65  | 58.65 | 59.44 |
| 69.00   | 58.48 | 58.59 | 58.65  | 58.65 | 59.44 |
| 70.00   | 58.48 | 58.59 | 58.65  | 58.65 | 59.48 |
| 71.00   | 58.48 | 58.59 | 58.64  | 58.65 | 59.48 |
| 72.00   | 58.41 | 58.51 | 58.55  | 58.61 | 58.66 |
| 73.00   | 58.41 | 58.50 | 58.55  | 58.61 | 58.66 |
| 74.00   | 58.45 | 58.55 | 58.59  | 58.61 | 59.02 |
| 75.00   | 58.52 | 58.68 | 58.78  | 58.78 | 59.02 |
| 76.00   | 58.60 | 58.79 | 58.88  | 58.97 | 59.09 |
| 77.00   | 58.60 | 58.80 | 58.88  | 58.99 | 59.09 |
| 78.00   | 58.60 | 58.78 | 58.87  | 58.97 | 59.07 |
| 79.00   | 58.60 | 58.80 | 58.88  | 58.99 | 59.09 |
| 80.00   | 58.41 | 58.50 | 58.55  | 58.59 | 58.66 |
| 81.00   | 58.41 | 58.50 | 58.55  | 58.59 | 58.66 |
| 82.00   | 58.45 | 58.50 | 58.55  | 58.61 | 58.91 |
| 83.00   | 58.54 | 58.62 | 58.70  | 58.78 | 58.91 |

18.3 L2 Requests (per normUnit)

| Channel | Min     | Q1      | Median  | Q3      | Max     |
|---------|---------|---------|---------|---------|---------|
| 0.00    | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3105.00 |
| 1.00    | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3104.00 |
| 2.00    | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3136.00 |
| 3.00    | 3096.00 | 3102.75 | 3108.00 | 3114.00 | 3137.00 |
| 4.00    | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 5.00    | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3136.00 |
| 6.00    | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 7.00    | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 8.00    | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3169.00 |
| 9.00    | 3088.00 | 3092.75 | 3096.00 | 3100.00 | 3169.00 |
| 10.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3168.00 |
| 11.00   | 3088.00 | 3092.75 | 3096.00 | 3100.00 | 3168.00 |
| 12.00   | 3088.00 | 3093.00 | 3097.00 | 3100.00 | 3168.00 |
| 13.00   | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3168.00 |
| 14.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3168.00 |
| 15.00   | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3169.00 |
| 16.00   | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3169.00 |
| 17.00   | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 18.00   | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3169.00 |
| 19.00   | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 20.00   | 3088.00 | 3096.00 | 3097.00 | 3100.00 | 3168.00 |
| 21.00   | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 22.00   | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 23.00   | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 24.00   | 3088.00 | 3095.25 | 3096.00 | 3100.00 | 3104.00 |
| 25.00   | 3088.00 | 3095.25 | 3096.00 | 3100.00 | 3105.00 |
| 26.00   | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3136.00 |
| 27.00   | 3096.00 | 3108.00 | 3108.00 | 3114.00 | 3136.00 |
| 28.00   | 3104.00 | 3118.25 | 3120.00 | 3128.00 | 3136.00 |
| 29.00   | 3104.00 | 3118.00 | 3120.00 | 3128.00 | 3136.00 |
| 30.00   | 3104.00 | 3118.25 | 3120.00 | 3128.00 | 3137.00 |
| 31.00   | 3104.00 | 3118.25 | 3120.00 | 3128.00 | 3137.00 |
| 32.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3105.00 |
| 33.00   | 3088.00 | 3093.00 | 3097.00 | 3100.00 | 3105.00 |
| 34.00   | 3088.00 | 3096.00 | 3100.00 | 3100.00 | 3128.00 |
| 35.00   | 3096.00 | 3108.00 | 3112.00 | 3114.00 | 3128.00 |
| 36.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 37.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3136.00 |
| 38.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 39.00   | 3104.00 | 3113.00 | 3121.00 | 3128.00 | 3137.00 |
| 40.00   | 3088.00 | 3096.00 | 3100.00 | 3100.00 | 3156.00 |
| 41.00   | 3088.00 | 3096.00 | 3100.00 | 3100.00 | 3157.00 |
| 42.00   | 3088.00 | 3096.00 | 3100.00 | 3100.00 | 3157.00 |
| 43.00   | 3088.00 | 3096.00 | 3100.00 | 3100.00 | 3156.00 |
| 44.00   | 3088.00 | 3096.00 | 3100.00 | 3101.00 | 3156.00 |
| 45.00   | 3088.00 | 3096.00 | 3100.00 | 3100.00 | 3156.00 |
| 46.00   | 3088.00 | 3096.00 | 3100.00 | 3100.00 | 3156.00 |
| 47.00   | 3088.00 | 3096.00 | 3100.00 | 3100.00 | 3157.00 |
| 48.00   | 3088.00 | 3096.00 | 3097.00 | 3101.75 | 3158.00 |
| 49.00   | 3088.00 | 3096.00 | 3096.50 | 3101.75 | 3156.00 |
| 50.00   | 3088.00 | 3096.00 | 3097.00 | 3101.75 | 3157.00 |
| 51.00   | 3088.00 | 3096.00 | 3097.00 | 3101.00 | 3156.00 |
| 52.00   | 3088.00 | 3096.00 | 3098.00 | 3102.50 | 3156.00 |
| 53.00   | 3088.00 | 3096.00 | 3097.00 | 3101.75 | 3156.00 |
| 54.00   | 3088.00 | 3096.00 | 3097.00 | 3101.75 | 3156.00 |
| 55.00   | 3088.00 | 3096.00 | 3097.00 | 3101.75 | 3156.00 |
| 56.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3104.00 |
| 57.00   | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3105.00 |
| 58.00   | 3088.00 | 3096.00 | 3096.00 | 3101.00 | 3128.00 |
| 59.00   | 3096.00 | 3108.00 | 3108.00 | 3114.00 | 3129.00 |
| 60.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 61.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 62.00   | 3104.00 | 3113.00 | 3120.00 | 3128.00 | 3136.00 |
| 63.00   | 3104.00 | 3113.00 | 3120.00 | 3128.00 | 3137.00 |
| 64.00   | 3088.00 | 3093.75 | 3096.00 | 3100.00 | 3158.00 |
| 65.00   | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3157.00 |
| 66.00   | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3156.00 |
| 67.00   | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3156.00 |
| 68.00   | 3088.00 | 3093.00 | 3097.00 | 3101.00 | 3156.00 |
| 69.00   | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3156.00 |
| 70.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3156.00 |
| 71.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3157.00 |
| 72.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3104.00 |
| 73.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3106.00 |
| 74.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3128.00 |
| 75.00   | 3096.00 | 3102.75 | 3108.50 | 3114.00 | 3128.00 |
| 76.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 77.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3136.00 |
| 78.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 79.00   | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3137.00 |
| 80.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3104.00 |
| 81.00   | 3088.00 | 3092.75 | 3096.00 | 3100.00 | 3105.00 |
| 82.00   | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3136.00 |
| 83.00   | 3096.00 | 3102.75 | 3108.00 | 3114.00 | 3136.00 |

|       |        |       |       |       |       |       |
|-------|--------|-------|-------|-------|-------|-------|
| 21. P | 84.00  | 58.60 | 58.78 | 58.87 | 58.97 | 59.09 |
|       | 85.00  | 58.60 | 58.78 | 58.88 | 58.99 | 59.09 |
| 22. P | 86.00  | 58.60 | 58.76 | 58.87 | 58.96 | 59.07 |
|       | 87.00  | 58.60 | 58.78 | 58.88 | 58.99 | 59.09 |
| 23. P | 88.00  | 58.48 | 58.54 | 58.59 | 58.65 | 59.29 |
|       | 89.00  | 58.47 | 58.54 | 58.59 | 58.65 | 59.29 |
| 24. P | 90.00  | 58.48 | 58.54 | 58.59 | 58.65 | 59.29 |
|       | 91.00  | 58.48 | 58.54 | 58.59 | 58.65 | 59.29 |
| 25. P | 92.00  | 58.47 | 58.54 | 58.59 | 58.65 | 59.29 |
|       | 93.00  | 58.48 | 58.54 | 58.59 | 58.65 | 59.29 |
| 26. P | 94.00  | 58.48 | 58.54 | 58.59 | 58.65 | 59.33 |
|       | 95.00  | 58.48 | 58.54 | 58.59 | 58.65 | 59.33 |
| 27. P | 96.00  | 58.48 | 58.59 | 58.61 | 58.68 | 59.44 |
|       | 97.00  | 58.48 | 58.59 | 58.61 | 58.70 | 59.44 |
| 28. P | 98.00  | 58.48 | 58.59 | 58.61 | 58.68 | 59.43 |
|       | 99.00  | 58.47 | 58.59 | 58.61 | 58.68 | 59.44 |
| 29. P | 100.00 | 58.48 | 58.59 | 58.61 | 58.68 | 59.44 |
|       | 101.00 | 58.47 | 58.59 | 58.61 | 58.68 | 59.44 |
| 30. P | 102.00 | 58.47 | 58.59 | 58.61 | 58.70 | 59.48 |
|       | 103.00 | 58.48 | 58.59 | 58.61 | 58.68 | 59.48 |
| 31. P | 104.00 | 58.45 | 58.52 | 58.55 | 58.61 | 58.66 |
|       | 105.00 | 58.45 | 58.52 | 58.55 | 58.61 | 58.66 |
| 32. P | 106.00 | 58.45 | 58.55 | 58.57 | 58.65 | 59.02 |
|       | 107.00 | 58.52 | 58.70 | 58.70 | 58.78 | 59.00 |
| 33. P | 108.00 | 58.65 | 58.81 | 58.88 | 58.99 | 59.09 |
|       | 109.00 | 58.67 | 58.81 | 58.88 | 58.99 | 59.09 |
| 34. P | 110.00 | 58.63 | 58.81 | 58.87 | 58.97 | 59.07 |
|       | 111.00 | 58.67 | 58.81 | 58.88 | 58.99 | 59.09 |
| 35. P | 112.00 | 58.41 | 58.50 | 58.55 | 58.55 | 58.66 |
|       | 113.00 | 58.41 | 58.50 | 58.55 | 58.55 | 58.66 |
| 36. P | 114.00 | 58.45 | 58.55 | 58.55 | 58.61 | 59.02 |
|       | 115.00 | 58.52 | 58.67 | 58.70 | 58.78 | 59.02 |
| 37. P | 116.00 | 58.58 | 58.78 | 58.88 | 58.88 | 59.09 |
|       | 117.00 | 58.58 | 58.78 | 58.88 | 58.88 | 59.09 |
| 38. P | 118.00 | 58.60 | 58.76 | 58.87 | 58.87 | 59.07 |
|       | 119.00 | 58.58 | 58.78 | 58.88 | 58.88 | 59.09 |
| 39. P | 120.00 | 58.48 | 58.59 | 58.59 | 58.65 | 59.44 |
|       | 121.00 | 58.47 | 58.59 | 58.59 | 58.65 | 59.44 |
| 40. P | 122.00 | 58.47 | 58.59 | 58.59 | 58.65 | 59.44 |
|       | 123.00 | 58.47 | 58.57 | 58.59 | 58.65 | 59.43 |
| 41. P | 124.00 | 58.48 | 58.59 | 58.59 | 58.65 | 59.44 |
|       | 125.00 | 58.48 | 58.57 | 58.59 | 58.65 | 59.43 |
| 42. P | 126.00 | 58.48 | 58.59 | 58.59 | 58.65 | 59.48 |
|       | 127.00 | 58.47 | 58.59 | 58.59 | 58.65 | 59.48 |

|        |         |         |         |         |         |
|--------|---------|---------|---------|---------|---------|
| 84.00  | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3136.00 |
| 85.00  | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3136.00 |
| 86.00  | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3136.00 |
| 87.00  | 3104.00 | 3113.00 | 3120.00 | 3128.00 | 3137.00 |
| 88.00  | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3168.00 |
| 89.00  | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3168.00 |
| 90.00  | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3168.00 |
| 91.00  | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3168.00 |
| 92.00  | 3088.00 | 3093.00 | 3096.00 | 3100.00 | 3168.00 |
| 93.00  | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3168.00 |
| 94.00  | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3168.00 |
| 95.00  | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3168.00 |
| 96.00  | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3158.00 |
| 97.00  | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3157.00 |
| 98.00  | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3157.00 |
| 99.00  | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3156.00 |
| 100.00 | 3088.00 | 3096.00 | 3097.00 | 3100.25 | 3156.00 |
| 101.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3156.00 |
| 102.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3156.00 |
| 103.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3157.00 |
| 104.00 | 3088.00 | 3092.00 | 3096.00 | 3100.00 | 3104.00 |
| 105.00 | 3088.00 | 3092.75 | 3096.00 | 3100.00 | 3105.00 |
| 106.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3128.00 |
| 107.00 | 3096.00 | 3108.00 | 3108.00 | 3114.00 | 3128.00 |
| 108.00 | 3104.00 | 3113.00 | 3120.00 | 3128.00 | 3136.00 |
| 109.00 | 3104.00 | 3112.00 | 3120.00 | 3128.00 | 3136.00 |
| 110.00 | 3104.00 | 3113.00 | 3120.00 | 3128.00 | 3137.00 |
| 111.00 | 3104.00 | 3113.00 | 3120.00 | 3128.00 | 3137.00 |
| 112.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3104.00 |
| 113.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3105.00 |
| 114.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3136.00 |
| 115.00 | 3096.00 | 3108.00 | 3108.00 | 3114.00 | 3136.00 |
| 116.00 | 3104.00 | 3120.00 | 3120.00 | 3128.00 | 3136.00 |
| 117.00 | 3104.00 | 3120.00 | 3120.00 | 3128.00 | 3136.00 |
| 118.00 | 3104.00 | 3120.00 | 3120.00 | 3128.00 | 3137.00 |
| 119.00 | 3104.00 | 3120.00 | 3120.00 | 3128.00 | 3137.00 |
| 120.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 121.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 122.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 123.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 124.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 125.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 126.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |
| 127.00 | 3088.00 | 3096.00 | 3096.00 | 3100.00 | 3168.00 |

18.4 L2 Requests (per normUnit)

| Channel | L2 Read | L2 Write | L2 Atomic |
|---------|---------|----------|-----------|
| 0.00    | 2099.13 | 1024.00  | 0.00      |
| 1.00    | 2099.15 | 1024.00  | 0.00      |
| 2.00    | 2099.13 | 1024.00  | 0.00      |
| 3.00    | 2099.17 | 1024.00  | 0.00      |
| 4.00    | 2078.89 | 1024.00  | 0.00      |
| 5.00    | 2078.49 | 1024.00  | 0.00      |
| 6.00    | 2088.84 | 1024.00  | 0.00      |
| 7.00    | 2078.49 | 1024.00  | 0.00      |
| 8.00    | 2075.23 | 1024.00  | 0.00      |
| 9.00    | 2075.21 | 1024.00  | 0.00      |
| 10.00   | 2075.26 | 1024.00  | 0.00      |
| 11.00   | 2075.36 | 1024.00  | 0.00      |
| 12.00   | 2075.23 | 1024.00  | 0.00      |
| 13.00   | 2075.26 | 1024.00  | 0.00      |
| 14.00   | 2071.95 | 1024.00  | 0.00      |
| 15.00   | 2071.97 | 1024.00  | 0.00      |
| 16.00   | 2075.36 | 1024.00  | 0.00      |
| 17.00   | 2075.34 | 1024.00  | 0.00      |
| 18.00   | 2075.39 | 1024.00  | 0.00      |
| 19.00   | 2075.49 | 1024.00  | 0.00      |
| 20.00   | 2075.36 | 1024.00  | 0.00      |
| 21.00   | 2075.39 | 1024.00  | 0.00      |
| 22.00   | 2072.27 | 1024.00  | 0.00      |
| 23.00   | 2072.29 | 1024.00  | 0.00      |
| 24.00   | 2099.58 | 1024.00  | 0.00      |
| 25.00   | 2099.60 | 1024.00  | 0.00      |
| 26.00   | 2099.58 | 1024.00  | 0.00      |
| 27.00   | 2099.62 | 1024.00  | 0.00      |
| 28.00   | 2078.84 | 1024.00  | 0.00      |
| 29.00   | 2078.43 | 1024.00  | 0.00      |
| 30.00   | 2089.04 | 1024.00  | 0.00      |
| 31.00   | 2078.43 | 1024.00  | 0.00      |
| 32.00   | 2100.26 | 1024.00  | 0.00      |
| 33.00   | 2100.28 | 1024.00  | 0.00      |
| 34.00   | 2100.26 | 1024.00  | 0.00      |
| 35.00   | 2100.30 | 1024.00  | 0.00      |
| 36.00   | 2079.58 | 1024.00  | 0.00      |
| 37.00   | 2079.17 | 1024.00  | 0.00      |
| 38.00   | 2089.75 | 1024.00  | 0.00      |
| 39.00   | 2079.17 | 1024.00  | 0.00      |
| 40.00   | 2075.84 | 1024.00  | 0.00      |
| 41.00   | 2075.82 | 1024.00  | 0.00      |
| 42.00   | 2075.86 | 1024.00  | 0.00      |
| 43.00   | 2075.96 | 1024.00  | 0.00      |
| 44.00   | 2075.84 | 1024.00  | 0.00      |
| 45.00   | 2075.86 | 1024.00  | 0.00      |
| 46.00   | 2072.48 | 1024.00  | 0.00      |
| 47.00   | 2072.50 | 1024.00  | 0.00      |
| 48.00   | 2075.36 | 1024.00  | 0.00      |

18.5 L2-Fabric Requests (per normUnit)

| Channel | L2-Fabric Read | L2-Fabric Write and Atomic | L2-Fabric Atomic |
|---------|----------------|----------------------------|------------------|
| 0.00    | 1156.03        | 256.00                     | 0.00             |
| 1.00    | 1156.19        | 256.00                     | 0.00             |
| 2.00    | 1156.53        | 256.00                     | 0.00             |
| 3.00    | 1158.36        | 256.00                     | 0.00             |
| 4.00    | 1160.05        | 256.00                     | 0.00             |
| 5.00    | 1160.03        | 256.00                     | 0.00             |
| 6.00    | 1160.07        | 256.00                     | 0.00             |
| 7.00    | 1160.23        | 256.00                     | 0.00             |
| 8.00    | 1157.17        | 256.00                     | 0.00             |
| 9.00    | 1157.13        | 256.00                     | 0.00             |
| 10.00   | 1157.11        | 256.00                     | 0.00             |
| 11.00   | 1157.10        | 256.00                     | 0.00             |
| 12.00   | 1157.34        | 256.00                     | 0.00             |
| 13.00   | 1157.10        | 256.00                     | 0.00             |
| 14.00   | 1157.09        | 256.00                     | 0.00             |
| 15.00   | 1157.10        | 256.00                     | 0.00             |
| 16.00   | 1157.17        | 256.00                     | 0.00             |
| 17.00   | 1157.13        | 256.00                     | 0.00             |
| 18.00   | 1157.11        | 256.00                     | 0.00             |
| 19.00   | 1157.10        | 256.00                     | 0.00             |
| 20.00   | 1157.34        | 256.00                     | 0.00             |
| 21.00   | 1157.10        | 256.00                     | 0.00             |
| 22.00   | 1157.09        | 256.00                     | 0.00             |
| 23.00   | 1157.10        | 256.00                     | 0.00             |
| 24.00   | 1156.03        | 256.00                     | 0.00             |
| 25.00   | 1156.19        | 256.00                     | 0.00             |
| 26.00   | 1156.53        | 256.00                     | 0.00             |
| 27.00   | 1158.36        | 256.00                     | 0.00             |
| 28.00   | 1160.05        | 256.00                     | 0.00             |
| 29.00   | 1160.03        | 256.00                     | 0.00             |
| 30.00   | 1160.07        | 256.00                     | 0.00             |
| 31.00   | 1160.23        | 256.00                     | 0.00             |
| 32.00   | 1156.03        | 256.00                     | 0.00             |
| 33.00   | 1156.19        | 256.00                     | 0.00             |
| 34.00   | 1156.53        | 256.00                     | 0.00             |
| 35.00   | 1158.36        | 256.00                     | 0.00             |
| 36.00   | 1160.05        | 256.00                     | 0.00             |
| 37.00   | 1160.03        | 256.00                     | 0.00             |
| 38.00   | 1160.07        | 256.00                     | 0.00             |
| 39.00   | 1160.23        | 256.00                     | 0.00             |
| 40.00   | 1157.17        | 256.00                     | 0.00             |
| 41.00   | 1157.13        | 256.00                     | 0.00             |
| 42.00   | 1157.11        | 256.00                     | 0.00             |
| 43.00   | 1157.10        | 256.00                     | 0.00             |
| 44.00   | 1157.34        | 256.00                     | 0.00             |
| 45.00   | 1157.10        | 256.00                     | 0.00             |
| 46.00   | 1157.09        | 256.00                     | 0.00             |
| 47.00   | 1157.10        | 256.00                     | 0.00             |
| 48.00   | 1157.17        | 256.00                     | 0.00             |

| Menu <span>⌵</span>       |  |         |         |      | TOP N: 10 |  |         |        |      | Kernels: <span>matmul_kernel</span> <span>⌵</span> |  |         |        |      | Report Bug |  |         |        |      |
|---------------------------|--|---------|---------|------|-----------|--|---------|--------|------|----------------------------------------------------|--|---------|--------|------|------------|--|---------|--------|------|
| NORMALIZATION: per_kernel |  |         |         |      | GCD: ALL  |  |         |        |      | DISPATCH FILTER: ALL                               |  |         |        |      |            |  |         |        |      |
| 51.00                     |  | 2075.49 | 1024.00 | 0.00 | 51.00     |  | 1157.10 | 256.00 | 0.00 | 51.00                                              |  | 1157.10 | 256.00 | 0.00 | 51.00      |  | 1157.10 | 256.00 | 0.00 |
| 52.00                     |  | 2075.36 | 1024.00 | 0.00 | 52.00     |  | 1157.34 | 256.00 | 0.00 | 52.00                                              |  | 1157.34 | 256.00 | 0.00 | 52.00      |  | 1157.34 | 256.00 | 0.00 |
| 53.00                     |  | 2075.39 | 1024.00 | 0.00 | 53.00     |  | 1157.10 | 256.00 | 0.00 | 53.00                                              |  | 1157.10 | 256.00 | 0.00 | 53.00      |  | 1157.10 | 256.00 | 0.00 |
| 54.00                     |  | 2072.40 | 1024.00 | 0.00 | 54.00     |  | 1157.09 | 256.00 | 0.00 | 54.00                                              |  | 1157.09 | 256.00 | 0.00 | 54.00      |  | 1157.09 | 256.00 | 0.00 |
| 55.00                     |  | 2072.42 | 1024.00 | 0.00 | 55.00     |  | 1157.10 | 256.00 | 0.00 | 55.00                                              |  | 1157.10 | 256.00 | 0.00 | 55.00      |  | 1157.10 | 256.00 | 0.00 |
| 56.00                     |  | 2099.71 | 1024.00 | 0.00 | 56.00     |  | 1156.03 | 256.00 | 0.00 | 56.00                                              |  | 1156.03 | 256.00 | 0.00 | 56.00      |  | 1156.03 | 256.00 | 0.00 |
| 57.00                     |  | 2099.73 | 1024.00 | 0.00 | 57.00     |  | 1156.19 | 256.00 | 0.00 | 57.00                                              |  | 1156.19 | 256.00 | 0.00 | 57.00      |  | 1156.19 | 256.00 | 0.00 |
| 58.00                     |  | 2099.71 | 1024.00 | 0.00 | 58.00     |  | 1156.53 | 256.00 | 0.00 | 58.00                                              |  | 1156.53 | 256.00 | 0.00 | 58.00      |  | 1156.53 | 256.00 | 0.00 |
| 59.00                     |  | 2099.75 | 1024.00 | 0.00 | 59.00     |  | 1158.36 | 256.00 | 0.00 | 59.00                                              |  | 1158.36 | 256.00 | 0.00 | 59.00      |  | 1158.36 | 256.00 | 0.00 |
| 60.00                     |  | 2078.71 | 1024.00 | 0.00 | 60.00     |  | 1160.05 | 256.00 | 0.00 | 60.00                                              |  | 1160.05 | 256.00 | 0.00 | 60.00      |  | 1160.05 | 256.00 | 0.00 |
| 61.00                     |  | 2078.30 | 1024.00 | 0.00 | 61.00     |  | 1160.03 | 256.00 | 0.00 | 61.00                                              |  | 1160.03 | 256.00 | 0.00 | 61.00      |  | 1160.03 | 256.00 | 0.00 |
| 62.00                     |  | 2089.04 | 1024.00 | 0.00 | 62.00     |  | 1160.07 | 256.00 | 0.00 | 62.00                                              |  | 1160.07 | 256.00 | 0.00 | 62.00      |  | 1160.07 | 256.00 | 0.00 |
| 63.00                     |  | 2078.30 | 1024.00 | 0.00 | 63.00     |  | 1160.23 | 256.00 | 0.00 | 63.00                                              |  | 1160.23 | 256.00 | 0.00 | 63.00      |  | 1160.23 | 256.00 | 0.00 |
| 64.00                     |  | 2075.26 | 1024.00 | 0.00 | 64.00     |  | 1157.17 | 256.00 | 0.00 | 64.00                                              |  | 1157.17 | 256.00 | 0.00 | 64.00      |  | 1157.17 | 256.00 | 0.00 |
| 65.00                     |  | 2075.24 | 1024.00 | 0.00 | 65.00     |  | 1157.13 | 256.00 | 0.00 | 65.00                                              |  | 1157.13 | 256.00 | 0.00 | 65.00      |  | 1157.13 | 256.00 | 0.00 |
| 66.00                     |  | 2075.28 | 1024.00 | 0.00 | 66.00     |  | 1157.11 | 256.00 | 0.00 | 66.00                                              |  | 1157.11 | 256.00 | 0.00 | 66.00      |  | 1157.11 | 256.00 | 0.00 |
| 67.00                     |  | 2075.38 | 1024.00 | 0.00 | 67.00     |  | 1157.10 | 256.00 | 0.00 | 67.00                                              |  | 1157.10 | 256.00 | 0.00 | 67.00      |  | 1157.10 | 256.00 | 0.00 |
| 68.00                     |  | 2075.26 | 1024.00 | 0.00 | 68.00     |  | 1157.34 | 256.00 | 0.00 | 68.00                                              |  | 1157.34 | 256.00 | 0.00 | 68.00      |  | 1157.34 | 256.00 | 0.00 |
| 69.00                     |  | 2075.28 | 1024.00 | 0.00 | 69.00     |  | 1157.10 | 256.00 | 0.00 | 69.00                                              |  | 1157.10 | 256.00 | 0.00 | 69.00      |  | 1157.10 | 256.00 | 0.00 |
| 70.00                     |  | 2072.19 | 1024.00 | 0.00 | 70.00     |  | 1157.09 | 256.00 | 0.00 | 70.00                                              |  | 1157.09 | 256.00 | 0.00 | 70.00      |  | 1157.09 | 256.00 | 0.00 |
| 71.00                     |  | 2072.21 | 1024.00 | 0.00 | 71.00     |  | 1157.10 | 256.00 | 0.00 | 71.00                                              |  | 1157.10 | 256.00 | 0.00 | 71.00      |  | 1157.10 | 256.00 | 0.00 |
| 72.00                     |  | 2099.39 | 1024.00 | 0.00 | 72.00     |  | 1156.03 | 256.00 | 0.00 | 72.00                                              |  | 1156.03 | 256.00 | 0.00 | 72.00      |  | 1156.03 | 256.00 | 0.00 |
| 73.00                     |  | 2099.41 | 1024.00 | 0.00 | 73.00     |  | 1156.19 | 256.00 | 0.00 | 73.00                                              |  | 1156.19 | 256.00 | 0.00 | 73.00      |  | 1156.19 | 256.00 | 0.00 |
| 74.00                     |  | 2099.39 | 1024.00 | 0.00 | 74.00     |  | 1156.53 | 256.00 | 0.00 | 74.00                                              |  | 1156.53 | 256.00 | 0.00 | 74.00      |  | 1156.53 | 256.00 | 0.00 |
| 75.00                     |  | 2099.43 | 1024.00 | 0.00 | 75.00     |  | 1158.36 | 256.00 | 0.00 | 75.00                                              |  | 1158.36 | 256.00 | 0.00 | 75.00      |  | 1158.36 | 256.00 | 0.00 |
| 76.00                     |  | 2078.71 | 1024.00 | 0.00 | 76.00     |  | 1160.05 | 256.00 | 0.00 | 76.00                                              |  | 1160.05 | 256.00 | 0.00 | 76.00      |  | 1160.05 | 256.00 | 0.00 |
| 77.00                     |  | 2078.30 | 1024.00 | 0.00 | 77.00     |  | 1160.03 | 256.00 | 0.00 | 77.00                                              |  | 1160.03 | 256.00 | 0.00 | 77.00      |  | 1160.03 | 256.00 | 0.00 |
| 78.00                     |  | 2088.88 | 1024.00 | 0.00 | 78.00     |  | 1160.07 | 256.00 | 0.00 | 78.00                                              |  | 1160.07 | 256.00 | 0.00 | 78.00      |  | 1160.07 | 256.00 | 0.00 |
| 79.00                     |  | 2078.30 | 1024.00 | 0.00 | 79.00     |  | 1160.23 | 256.00 | 0.00 | 79.00                                              |  | 1160.23 | 256.00 | 0.00 | 79.00      |  | 1160.23 | 256.00 | 0.00 |
| 80.00                     |  | 2100.05 | 1024.00 | 0.00 | 80.00     |  | 1156.03 | 256.00 | 0.00 | 80.00                                              |  | 1156.03 | 256.00 | 0.00 | 80.00      |  | 1156.03 | 256.00 | 0.00 |
| 81.00                     |  | 2100.07 | 1024.00 | 0.00 | 81.00     |  | 1156.19 | 256.00 | 0.00 | 81.00                                              |  | 1156.19 | 256.00 | 0.00 | 81.00      |  | 1156.19 | 256.00 | 0.00 |
| 82.00                     |  | 2100.05 | 1024.00 | 0.00 | 82.00     |  | 1156.53 | 256.00 | 0.00 | 82.00                                              |  | 1156.53 | 256.00 | 0.00 | 82.00      |  | 1156.53 | 256.00 | 0.00 |
| 83.00                     |  | 2100.09 | 1024.00 | 0.00 | 83.00     |  | 1158.36 | 256.00 | 0.00 | 83.00                                              |  | 1158.36 | 256.00 | 0.00 | 83.00      |  | 1158.36 | 256.00 | 0.00 |
| 84.00                     |  | 2079.16 | 1024.00 | 0.00 | 84.00     |  | 1160.05 | 256.00 | 0.00 | 84.00                                              |  | 1160.05 | 256.00 | 0.00 | 84.00      |  | 1160.05 | 256.00 | 0.00 |
| 85.00                     |  | 2078.75 | 1024.00 | 0.00 | 85.00     |  | 1160.03 | 256.00 | 0.00 | 85.00                                              |  | 1160.03 | 256.00 | 0.00 | 85.00      |  | 1160.03 | 256.00 | 0.00 |
| 86.00                     |  | 2089.43 | 1024.00 | 0.00 | 86.00     |  | 1160.07 | 256.00 | 0.00 | 86.00                                              |  | 1160.07 | 256.00 | 0.00 | 86.00      |  | 1160.07 | 256.00 | 0.00 |
| 87.00                     |  | 2078.75 | 1024.00 | 0.00 | 87.00     |  | 1160.23 | 256.00 | 0.00 | 87.00                                              |  | 1160.23 | 256.00 | 0.00 | 87.00      |  | 1160.23 | 256.00 | 0.00 |
| 88.00                     |  | 2075.63 | 1024.00 | 0.00 | 88.00     |  | 1157.17 | 256.00 | 0.00 | 88.00                                              |  | 1157.17 | 256.00 | 0.00 | 88.00      |  | 1157.17 | 256.00 | 0.00 |
| 89.00                     |  | 2075.61 | 1024.00 | 0.00 | 89.00     |  | 1157.13 | 256.00 | 0.00 | 89.00                                              |  | 1157.13 | 256.00 | 0.00 | 89.00      |  | 1157.13 | 256.00 | 0.00 |
| 90.00                     |  | 2075.65 | 1024.00 | 0.00 | 90.00     |  | 1157.11 | 256.00 | 0.00 | 90.00                                              |  | 1157.11 | 256.00 | 0.00 | 90.00      |  | 1157.11 | 256.00 | 0.00 |
| 91.00                     |  | 2075.75 | 1024.00 | 0.00 | 91.00     |  | 1157.10 | 256.00 | 0.00 | 91.00                                              |  | 1157.10 | 256.00 | 0.00 | 91.00      |  | 1157.10 | 256.00 | 0.00 |
| 92.00                     |  | 2075.63 | 1024.00 | 0.00 | 92.00     |  | 1157.34 | 256.00 | 0.00 | 92.00                                              |  | 1157.34 | 256.00 | 0.00 | 92.00      |  | 1157.34 | 256.00 | 0.00 |
| 93.00                     |  | 2075.65 | 1024.00 | 0.00 | 93.00     |  | 1157.10 | 256.00 | 0.00 | 93.00                                              |  | 1157.10 | 256.00 | 0.00 | 93.00      |  | 1157.10 | 256.00 | 0.00 |
| 94.00                     |  | 2072.48 | 1024.00 | 0.00 | 94.00     |  | 1157.09 | 256.00 | 0.00 | 94.00                                              |  | 1157.09 | 256.00 | 0.00 | 94.00      |  | 1157.09 | 256.00 | 0.00 |
| 95.00                     |  | 2072.50 | 1024.00 | 0.00 | 95.00     |  | 1157.10 | 256.00 | 0.00 | 95.00                                              |  | 1157.10 | 256.00 | 0.00 | 95.00      |  | 1157.10 | 256.00 | 0.00 |
| 96.00                     |  | 2076.20 | 1024.00 | 0.00 | 96.00     |  | 1157.17 | 256.00 | 0.00 | 96.00                                              |  | 1157.17 | 256.00 | 0.00 | 96.00      |  | 1157.17 | 256.00 | 0.00 |
| 97.00                     |  | 2076.18 | 1024.00 | 0.00 | 97.00     |  | 1157.13 | 256.00 | 0.00 | 97.00                                              |  | 1157.13 | 256.00 | 0.00 | 97.00      |  | 1157.13 | 256.00 | 0.00 |
| 98.00                     |  | 2076.23 | 1024.00 | 0.00 | 98.00     |  | 1157.11 | 256.00 | 0.00 | 98.00                                              |  | 1157.11 | 256.00 | 0.00 | 98.00      |  | 1157.11 | 256.00 | 0.00 |
| 99.00                     |  | 2076.33 | 1024.00 | 0.00 | 99.00     |  | 1157.10 | 256.00 | 0.00 | 99.00                                              |  | 1157.10 | 256.00 | 0.00 | 99.00      |  | 1157.10 | 256.00 | 0.00 |
| 100.00                    |  | 2076.20 | 1024.00 | 0.00 | 100.00    |  | 1157.34 | 256.00 | 0.00 | 100.00                                             |  | 1157.34 | 256.00 | 0.00 | 100.00     |  | 1157.34 | 256.00 | 0.00 |
| 101.00                    |  | 2076.23 | 1024.00 | 0.00 | 101.00    |  | 1157.10 | 256.00 | 0.00 | 101.00                                             |  | 1157.10 | 256.00 | 0.00 | 101.00     |  | 1157.10 | 256.00 | 0.00 |
| 102.00                    |  | 2072.88 | 1024.00 | 0.00 | 102.00    |  | 1157.09 | 256.00 | 0.00 | 102.00                                             |  | 1157.09 | 256.00 | 0.00 | 102.00     |  | 1157.09 | 256.00 | 0.00 |
| 103.00                    |  | 2072.89 | 1024.00 | 0.00 | 103.00    |  | 1157.10 | 256.00 | 0.00 | 103.00                                             |  | 1157.10 | 256.00 | 0.00 | 103.00     |  | 1157.10 | 256.00 | 0.00 |
| 104.00                    |  | 2101.03 | 1024.00 | 0.00 | 104.00    |  | 1156.03 | 256.00 | 0.00 | 104.00                                             |  | 1156.03 | 256.00 | 0.00 | 104.00     |  | 1156.03 | 256.00 | 0.00 |
| 105.00                    |  | 2101.05 | 1024.00 | 0.00 | 105.00    |  | 1156.19 | 256.00 | 0.00 | 105.00                                             |  | 1156.19 | 256.00 | 0.00 | 105.00     |  | 1156.19 | 256.00 | 0.00 |
| 106.00                    |  | 2101.03 | 1024.00 | 0.00 | 106.00    |  | 1156.53 | 256.00 | 0.00 | 106.00                                             |  | 1156.53 | 256.00 | 0.00 | 106.00     |  | 1156.53 | 256.00 | 0.00 |
| 107.00                    |  | 2101.07 | 1024.00 | 0.00 | 107.00    |  | 1158.36 | 256.00 | 0.00 | 107.00                                             |  | 1158.36 | 256.00 | 0.00 | 107.00     |  | 1158.36 | 256.00 | 0.00 |
| 108.00                    |  | 2079.92 | 1024.00 | 0.00 | 108.00    |  | 1160.05 | 256.00 | 0.00 | 108.00                                             |  | 1160.05 | 256.00 | 0.00 | 108.00     |  | 1160.05 | 256.00 | 0.00 |
| 109.00                    |  | 2079.51 | 1024.00 | 0.00 | 109.00    |  | 1160.03 | 256.00 | 0.00 | 109.00                                             |  | 1160.03 | 256.00 | 0.00 | 109.00     |  | 1160.03 | 256.00 | 0.00 |
| 110.00                    |  | 2090.30 | 1024.00 | 0.00 | 110.00    |  | 1160.07 | 256.00 | 0.00 | 110.00                                             |  | 1160.07 | 256.00 | 0.00 | 110.00     |  | 1160.07 | 256.00 | 0.00 |
| 111.00                    |  | 2079.51 | 1024.00 | 0.00 | 111.00    |  | 1160.23 | 256.00 | 0.00 | 111.00                                             |  | 1160.23 | 256.00 | 0.00 | 111.00     |  | 1160.23 | 256.00 | 0.00 |
| 112.00                    |  | 2098.53 | 1024.00 | 0.00 | 112.00    |  | 1156.03 | 256.00 | 0.00 | 112.00                                             |  | 1156.03 | 256.00 | 0.00 | 112.00     |  | 1156.03 | 256.00 | 0.00 |
| 113.00                    |  | 2098.55 | 1024.00 | 0.00 | 113.00    |  | 1156.19 | 256.00 | 0.00 | 113.00                                             |  | 1156.19 | 256.00 | 0.00 | 113.00     |  |         |        |      |

| Menu             |        |        |        |        |        |  |  |  |  | Report Bug    |        |        |        |        |        |  |  |  |  |
|------------------|--------|--------|--------|--------|--------|--|--|--|--|---------------|--------|--------|--------|--------|--------|--|--|--|--|
| NORMALIZATION:   |        |        |        |        |        |  |  |  |  | Kernels:      |        |        |        |        |        |  |  |  |  |
| per_kernel       |        |        |        |        |        |  |  |  |  | matmul_kernel |        |        |        |        |        |  |  |  |  |
| GCD: ALL         |        |        |        |        |        |  |  |  |  | ALL           |        |        |        |        |        |  |  |  |  |
| DISPATCH FILTER: |        |        |        |        |        |  |  |  |  | ALL           |        |        |        |        |        |  |  |  |  |
| TOP N:           |        |        |        |        |        |  |  |  |  | 10            |        |        |        |        |        |  |  |  |  |
| 18.00            | 494.89 | 533.17 | 571.08 | 580.36 | 725.70 |  |  |  |  | 18.00         | 308.53 | 312.78 | 315.29 | 358.47 | 483.96 |  |  |  |  |
| 19.00            | 497.29 | 540.77 | 575.46 | 586.16 | 749.38 |  |  |  |  | 19.00         | 317.52 | 320.89 | 323.84 | 363.57 | 504.42 |  |  |  |  |
| 20.00            | 499.89 | 541.41 | 577.04 | 583.33 | 738.70 |  |  |  |  | 20.00         | 311.44 | 315.02 | 317.66 | 350.45 | 487.51 |  |  |  |  |
| 21.00            | 496.87 | 538.87 | 571.56 | 581.78 | 748.01 |  |  |  |  | 21.00         | 313.41 | 316.95 | 319.62 | 354.43 | 490.67 |  |  |  |  |
| 22.00            | 520.33 | 562.07 | 600.13 | 608.49 | 746.40 |  |  |  |  | 22.00         | 319.34 | 323.46 | 326.78 | 356.41 | 491.17 |  |  |  |  |
| 23.00            | 500.61 | 545.92 | 581.38 | 589.33 | 754.10 |  |  |  |  | 23.00         | 314.34 | 318.21 | 320.92 | 368.73 | 504.75 |  |  |  |  |
| 24.00            | 479.38 | 514.07 | 544.85 | 553.29 | 695.14 |  |  |  |  | 24.00         | 303.69 | 308.10 | 310.59 | 340.60 | 471.77 |  |  |  |  |
| 25.00            | 487.58 | 533.69 | 558.12 | 566.20 | 732.34 |  |  |  |  | 25.00         | 310.37 | 313.85 | 316.17 | 353.10 | 490.47 |  |  |  |  |
| 26.00            | 500.16 | 541.09 | 569.46 | 578.86 | 741.97 |  |  |  |  | 26.00         | 316.55 | 321.37 | 323.79 | 362.38 | 500.53 |  |  |  |  |
| 27.00            | 496.67 | 542.85 | 570.01 | 578.51 | 734.87 |  |  |  |  | 27.00         | 315.70 | 320.44 | 322.99 | 360.31 | 498.87 |  |  |  |  |
| 28.00            | 499.82 | 536.13 | 570.75 | 579.23 | 743.37 |  |  |  |  | 28.00         | 314.27 | 317.04 | 320.00 | 352.26 | 489.09 |  |  |  |  |
| 29.00            | 490.31 | 536.17 | 566.70 | 575.69 | 743.21 |  |  |  |  | 29.00         | 315.80 | 318.40 | 321.33 | 353.04 | 490.08 |  |  |  |  |
| 30.00            | 531.85 | 570.77 | 601.62 | 611.74 | 748.65 |  |  |  |  | 30.00         | 326.91 | 330.65 | 333.75 | 372.68 | 514.98 |  |  |  |  |
| 31.00            | 510.21 | 553.67 | 585.84 | 594.92 | 753.08 |  |  |  |  | 31.00         | 322.74 | 326.59 | 329.20 | 375.76 | 518.14 |  |  |  |  |
| 32.00            | 480.55 | 507.32 | 547.72 | 558.43 | 713.25 |  |  |  |  | 32.00         | 305.13 | 308.51 | 311.04 | 338.35 | 471.05 |  |  |  |  |
| 33.00            | 497.54 | 526.52 | 562.89 | 573.55 | 744.62 |  |  |  |  | 33.00         | 314.82 | 318.00 | 320.85 | 357.86 | 496.13 |  |  |  |  |
| 34.00            | 493.52 | 521.06 | 563.88 | 572.94 | 734.28 |  |  |  |  | 34.00         | 309.96 | 313.62 | 316.54 | 345.33 | 478.81 |  |  |  |  |
| 35.00            | 500.03 | 527.75 | 569.17 | 578.89 | 730.81 |  |  |  |  | 35.00         | 318.75 | 321.78 | 324.03 | 359.72 | 496.63 |  |  |  |  |
| 36.00            | 498.39 | 525.08 | 568.60 | 581.58 | 739.37 |  |  |  |  | 36.00         | 312.33 | 315.53 | 318.27 | 349.50 | 482.74 |  |  |  |  |
| 37.00            | 496.08 | 522.89 | 568.14 | 578.37 | 757.54 |  |  |  |  | 37.00         | 315.13 | 317.50 | 320.45 | 351.21 | 486.95 |  |  |  |  |
| 38.00            | 520.48 | 546.07 | 594.20 | 604.51 | 769.40 |  |  |  |  | 38.00         | 321.92 | 324.80 | 328.14 | 362.55 | 501.50 |  |  |  |  |
| 39.00            | 503.49 | 533.30 | 575.83 | 584.79 | 735.68 |  |  |  |  | 39.00         | 316.04 | 319.24 | 322.35 | 348.71 | 483.29 |  |  |  |  |
| 40.00            | 473.29 | 500.35 | 545.04 | 560.76 | 705.63 |  |  |  |  | 40.00         | 305.80 | 308.78 | 310.24 | 334.66 | 457.64 |  |  |  |  |
| 41.00            | 489.62 | 515.80 | 562.06 | 574.27 | 709.20 |  |  |  |  | 41.00         | 311.85 | 314.07 | 316.25 | 349.46 | 475.30 |  |  |  |  |
| 42.00            | 496.63 | 523.20 | 572.15 | 585.23 | 758.07 |  |  |  |  | 42.00         | 318.63 | 321.63 | 323.26 | 351.92 | 482.14 |  |  |  |  |
| 43.00            | 497.26 | 522.69 | 572.74 | 585.62 | 738.55 |  |  |  |  | 43.00         | 317.79 | 320.89 | 322.98 | 351.91 | 480.29 |  |  |  |  |
| 44.00            | 499.37 | 527.80 | 572.85 | 590.43 | 744.92 |  |  |  |  | 44.00         | 314.21 | 316.77 | 319.14 | 348.53 | 475.57 |  |  |  |  |
| 45.00            | 497.91 | 526.11 | 571.12 | 586.87 | 745.94 |  |  |  |  | 45.00         | 316.57 | 319.10 | 321.63 | 350.01 | 476.05 |  |  |  |  |
| 46.00            | 529.98 | 557.38 | 606.06 | 617.37 | 799.35 |  |  |  |  | 46.00         | 328.39 | 331.23 | 334.34 | 366.72 | 499.18 |  |  |  |  |
| 47.00            | 517.44 | 543.13 | 592.63 | 605.94 | 736.22 |  |  |  |  | 47.00         | 323.68 | 326.47 | 329.54 | 364.99 | 496.80 |  |  |  |  |
| 48.00            | 467.19 | 514.80 | 548.55 | 559.87 | 687.46 |  |  |  |  | 48.00         | 305.07 | 307.22 | 309.92 | 330.39 | 443.54 |  |  |  |  |
| 49.00            | 481.26 | 530.35 | 563.91 | 573.74 | 714.97 |  |  |  |  | 49.00         | 311.16 | 313.23 | 315.90 | 342.52 | 452.91 |  |  |  |  |
| 50.00            | 494.95 | 533.95 | 570.38 | 580.67 | 703.22 |  |  |  |  | 50.00         | 316.23 | 318.92 | 321.50 | 346.28 | 461.80 |  |  |  |  |
| 51.00            | 492.30 | 539.51 | 571.93 | 581.98 | 725.30 |  |  |  |  | 51.00         | 317.57 | 319.59 | 322.36 | 347.07 | 462.20 |  |  |  |  |
| 52.00            | 495.34 | 540.73 | 574.18 | 585.55 | 715.35 |  |  |  |  | 52.00         | 312.42 | 314.87 | 317.52 | 344.11 | 454.32 |  |  |  |  |
| 53.00            | 492.13 | 539.11 | 573.95 | 584.76 | 725.40 |  |  |  |  | 53.00         | 314.02 | 316.35 | 319.47 | 345.51 | 457.48 |  |  |  |  |
| 54.00            | 528.00 | 569.81 | 606.21 | 616.19 | 742.81 |  |  |  |  | 54.00         | 327.26 | 329.50 | 332.35 | 360.51 | 474.73 |  |  |  |  |
| 55.00            | 511.14 | 555.13 | 590.84 | 602.05 | 741.70 |  |  |  |  | 55.00         | 321.41 | 324.10 | 327.50 | 357.93 | 470.82 |  |  |  |  |
| 56.00            | 475.31 | 517.91 | 547.79 | 557.00 | 695.71 |  |  |  |  | 56.00         | 305.62 | 307.77 | 310.23 | 329.79 | 446.69 |  |  |  |  |
| 57.00            | 493.34 | 538.89 | 566.90 | 573.72 | 725.03 |  |  |  |  | 57.00         | 314.96 | 316.73 | 319.89 | 345.67 | 465.24 |  |  |  |  |
| 58.00            | 492.10 | 533.50 | 561.85 | 568.00 | 698.07 |  |  |  |  | 58.00         | 311.16 | 313.01 | 316.51 | 336.57 | 456.18 |  |  |  |  |
| 59.00            | 496.89 | 538.47 | 567.64 | 577.23 | 736.89 |  |  |  |  | 59.00         | 318.40 | 320.25 | 323.18 | 348.44 | 467.34 |  |  |  |  |
| 60.00            | 495.04 | 537.82 | 570.00 | 579.82 | 717.43 |  |  |  |  | 60.00         | 312.76 | 314.52 | 317.83 | 338.08 | 454.66 |  |  |  |  |
| 61.00            | 490.83 | 536.83 | 568.15 | 576.56 | 717.96 |  |  |  |  | 61.00         | 314.32 | 316.51 | 319.78 | 340.49 | 459.65 |  |  |  |  |
| 62.00            | 520.93 | 560.36 | 591.44 | 601.83 | 730.37 |  |  |  |  | 62.00         | 321.54 | 324.10 | 328.04 | 352.54 | 471.36 |  |  |  |  |
| 63.00            | 498.16 | 544.35 | 575.60 | 584.48 | 743.06 |  |  |  |  | 63.00         | 316.05 | 318.50 | 322.09 | 341.27 | 460.54 |  |  |  |  |
| 64.00            | 477.21 | 513.26 | 549.29 | 562.86 | 711.22 |  |  |  |  | 64.00         | 303.91 | 307.20 | 310.33 | 329.96 | 460.86 |  |  |  |  |
| 65.00            | 492.81 | 530.59 | 565.04 | 575.60 | 708.55 |  |  |  |  | 65.00         | 310.64 | 314.04 | 316.43 | 341.20 | 480.00 |  |  |  |  |
| 66.00            | 509.66 | 535.62 | 572.19 | 586.05 | 713.12 |  |  |  |  | 66.00         | 316.09 | 319.21 | 321.78 | 347.82 | 483.35 |  |  |  |  |
| 67.00            | 507.28 | 537.08 | 571.82 | 584.39 | 719.05 |  |  |  |  | 67.00         | 316.42 | 320.13 | 322.58 | 347.53 | 482.30 |  |  |  |  |
| 68.00            | 502.93 | 531.13 | 568.73 | 582.29 | 725.20 |  |  |  |  | 68.00         | 311.56 | 314.12 | 316.64 | 338.99 | 472.46 |  |  |  |  |
| 69.00            | 499.30 | 531.48 | 566.71 | 577.07 | 724.65 |  |  |  |  | 69.00         | 313.02 | 315.60 | 318.65 | 341.30 | 477.94 |  |  |  |  |
| 70.00            | 538.72 | 567.95 | 600.25 | 615.50 | 742.80 |  |  |  |  | 70.00         | 326.61 | 329.61 | 331.82 | 358.28 | 500.83 |  |  |  |  |
| 71.00            | 521.57 | 548.68 | 582.76 | 599.17 | 741.13 |  |  |  |  | 71.00         | 322.17 | 325.35 | 327.79 | 356.65 | 496.63 |  |  |  |  |
| 72.00            | 485.33 | 511.72 | 542.36 | 550.30 | 701.90 |  |  |  |  | 72.00         | 304.55 | 307.45 | 309.03 | 332.97 | 463.82 |  |  |  |  |
| 73.00            | 506.27 | 538.22 | 558.78 | 567.62 | 738.04 |  |  |  |  | 73.00         | 313.09 | 316.95 | 318.54 | 349.23 | 482.42 |  |  |  |  |
| 74.00            | 500.84 | 530.27 | 557.56 | 564.72 | 727.01 |  |  |  |  | 74.00         | 310.42 | 313.43 | 315.22 | 340.31 | 471.28 |  |  |  |  |
| 75.00            | 508.56 | 534.55 | 561.87 | 569.29 | 737.63 |  |  |  |  | 75.00         | 317.52 | 320.51 | 322.42 | 352.52 | 485.98 |  |  |  |  |
| 76.00            | 504.02 | 531.21 | 564.87 | 575.39 | 709.83 |  |  |  |  | 76.00         | 311.23 | 314.25 | 316.17 | 342.07 | 471.24 |  |  |  |  |
| 77.00            | 500.72 | 531.65 | 561.95 | 571.46 | 710.82 |  |  |  |  | 77.00         | 313.56 | 316.61 | 318.44 | 345.09 | 479.03 |  |  |  |  |
| 78.00            | 529.03 | 555.93 | 588.19 | 598.20 | 736.30 |  |  |  |  | 78.00         | 322.40 | 325.44 | 327.23 | 355.91 | 489.73 |  |  |  |  |
| 79.00            | 510.35 | 537.76 | 571.28 | 578.83 | 733.56 |  |  |  |  | 79.00         | 315.84 | 319.49 | 321.71 | 345.33 | 476.55 |  |  |  |  |
| 80.00            | 487.60 | 526.93 | 549.11 | 555.59 | 696.52 |  |  |  |  | 80.00         | 305.13 | 307.25 | 309.30 | 338.00 | 476.06 |  |  |  |  |
| 81.00            | 507.21 | 545.35 | 564.68 | 571.56 | 704.45 |  |  |  |  | 81.00         | 315.44 | 317.79 | 319.71 | 356.21 | 509.41 |  |  |  |  |
| 82.00            | 499.49 | 527.35 | 561.44 | 569.96 | 739.82 |  |  |  |  | 82.00         | 311.10 | 313.59 | 315.74 | 345.09 | 486.85 |  |  |  |  |
| 83.00            | 506.00 | 535.60 | 568.44 | 575.84 | 721.26 |  |  |  |  | 83.00         | 319.83 | 322.00 | 323.78 | 358.48 | 511.38 |  |  |  |  |
| 84.00            | 500.64 | 543.97 | 567.96 | 575.34 | 715.48 |  |  |  |  | 84.00         | 312.56 | 315.20 | 316.74 | 347.46 | 489.93 |  |  |  |  |
| 85.00            | 500.39 | 542.32 | 563.54 | 571.77 | 709.41 |  |  |  |  | 85.00         | 315.23 | 317.25 | 319.02 | 351    |        |  |  |  |  |

|        |        |                           |        |        |        |          |                      |           |                        |        |            |
|--------|--------|---------------------------|--------|--------|--------|----------|----------------------|-----------|------------------------|--------|------------|
| Menu   | 0.00   | NORMALIZATION: per_kernel |        |        |        | GCD: ALL | DISPATCH FILTER: ALL | TOP N: 10 | KERNELS: matmul_kernel | 346.86 | Report Bug |
| 117.00 | 499.51 | 521.93                    | 570.22 | 583.37 | 576.45 | 725.34   | 346.86               | 510.56    |                        |        |            |
| 118.00 | 531.93 | 570.22                    | 583.37 | 603.88 | 610.88 | 754.14   | 346.86               | 510.56    |                        |        |            |
| 119.00 | 516.68 | 556.14                    | 584.57 | 593.62 | 593.62 | 725.34   | 346.86               | 514.88    |                        |        |            |
| 120.00 | 474.57 | 522.98                    | 551.41 | 560.61 | 560.61 | 688.30   | 346.86               | 460.54    |                        |        |            |
| 121.00 | 499.46 | 544.52                    | 571.57 | 581.70 | 581.70 | 711.13   | 346.86               | 482.51    |                        |        |            |
| 122.00 | 496.59 | 531.98                    | 569.49 | 576.94 | 576.94 | 710.56   | 346.86               | 471.11    |                        |        |            |
| 123.00 | 582.00 | 540.52                    | 575.71 | 584.12 | 584.12 | 702.31   | 346.86               | 489.38    |                        |        |            |
| 124.00 | 498.75 | 543.86                    | 572.97 | 580.63 | 580.63 | 704.98   | 346.86               | 475.82    |                        |        |            |
| 125.00 | 581.70 | 540.92                    | 568.44 | 577.94 | 577.94 | 707.35   | 346.86               | 475.86    |                        |        |            |
| 126.00 | 521.29 | 559.80                    | 596.50 | 603.74 | 603.74 | 738.63   | 346.86               | 478.69    |                        |        |            |
| 127.00 | 509.25 | 544.43                    | 580.29 | 589.09 | 589.09 | 703.02   | 346.86               | 485.32    |                        |        |            |

18.8 L2-Fabric Atomic Latency (Cycles)

| Channel | Min  | Q1   | Median | Q3   | Max  |
|---------|------|------|--------|------|------|
| 0.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 1.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 2.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 3.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 4.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 5.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 6.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 7.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 8.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 9.00    | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 10.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 11.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 12.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 13.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 14.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 15.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 16.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 17.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 18.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 19.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 20.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 21.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 22.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 23.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 24.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 25.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 26.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 27.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 28.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 29.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 30.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 31.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 32.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 33.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 34.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 35.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 36.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 37.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 38.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 39.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 40.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 41.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 42.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 43.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 44.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 45.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 46.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 47.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 48.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 49.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 50.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 51.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 52.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 53.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 54.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 55.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 56.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 57.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 58.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 59.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 60.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 61.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 62.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 63.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 64.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 65.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 66.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 67.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 68.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 69.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 70.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 71.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 72.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 73.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 74.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 75.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 76.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 77.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 78.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 79.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 80.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 81.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |
| 82.00   | 0.00 | 0.00 | 0.00   | 0.00 | 0.00 |

18.9 L2-Fabric Read Stall (Cycles per normUnit)

| Channel | L2-Fabric Read Stall (PCIe) | L2-Fabric Read Stall (Infinity Fabric™) | L2-Fabric Read Stall (HBM) |
|---------|-----------------------------|-----------------------------------------|----------------------------|
| 0.00    |                             |                                         |                            |
| 1.00    |                             |                                         |                            |
| 2.00    |                             |                                         |                            |
| 3.00    |                             |                                         |                            |
| 4.00    |                             |                                         |                            |
| 5.00    |                             |                                         |                            |
| 6.00    |                             |                                         |                            |
| 7.00    |                             |                                         |                            |
| 8.00    |                             |                                         |                            |
| 9.00    |                             |                                         |                            |
| 10.00   |                             |                                         |                            |
| 11.00   |                             |                                         |                            |
| 12.00   |                             |                                         |                            |
| 13.00   |                             |                                         |                            |
| 14.00   |                             |                                         |                            |
| 15.00   |                             |                                         |                            |
| 16.00   |                             |                                         |                            |
| 17.00   |                             |                                         |                            |
| 18.00   |                             |                                         |                            |
| 19.00   |                             |                                         |                            |
| 20.00   |                             |                                         |                            |
| 21.00   |                             |                                         |                            |
| 22.00   |                             |                                         |                            |
| 23.00   |                             |                                         |                            |
| 24.00   |                             |                                         |                            |
| 25.00   |                             |                                         |                            |
| 26.00   |                             |                                         |                            |
| 27.00   |                             |                                         |                            |
| 28.00   |                             |                                         |                            |
| 29.00   |                             |                                         |                            |
| 30.00   |                             |                                         |                            |
| 31.00   |                             |                                         |                            |
| 32.00   |                             |                                         |                            |
| 33.00   |                             |                                         |                            |
| 34.00   |                             |                                         |                            |
| 35.00   |                             |                                         |                            |
| 36.00   |                             |                                         |                            |
| 37.00   |                             |                                         |                            |
| 38.00   |                             |                                         |                            |
| 39.00   |                             |                                         |                            |
| 40.00   |                             |                                         |                            |
| 41.00   |                             |                                         |                            |
| 42.00   |                             |                                         |                            |
| 43.00   |                             |                                         |                            |
| 44.00   |                             |                                         |                            |
| 45.00   |                             |                                         |                            |
| 46.00   |                             |                                         |                            |
| 47.00   |                             |                                         |                            |
| 48.00   |                             |                                         |                            |
| 49.00   |                             |                                         |                            |
| 50.00   |                             |                                         |                            |
| 51.00   |                             |                                         |                            |
| 52.00   |                             |                                         |                            |
| 53.00   |                             |                                         |                            |
| 54.00   |                             |                                         |                            |
| 55.00   |                             |                                         |                            |
| 56.00   |                             |                                         |                            |
| 57.00   |                             |                                         |                            |
| 58.00   |                             |                                         |                            |
| 59.00   |                             |                                         |                            |
| 60.00   |                             |                                         |                            |
| 61.00   |                             |                                         |                            |
| 62.00   |                             |                                         |                            |
| 63.00   |                             |                                         |                            |
| 64.00   |                             |                                         |                            |
| 65.00   |                             |                                         |                            |
| 66.00   |                             |                                         |                            |
| 67.00   |                             |                                         |                            |
| 68.00   |                             |                                         |                            |
| 69.00   |                             |                                         |                            |
| 70.00   |                             |                                         |                            |
| 71.00   |                             |                                         |                            |
| 72.00   |                             |                                         |                            |
| 73.00   |                             |                                         |                            |
| 74.00   |                             |                                         |                            |
| 75.00   |                             |                                         |                            |
| 76.00   |                             |                                         |                            |
| 77.00   |                             |                                         |                            |
| 78.00   |                             |                                         |                            |
| 79.00   |                             |                                         |                            |
| 80.00   |                             |                                         |                            |
| 81.00   |                             |                                         |                            |
| 82.00   |                             |                                         |                            |



|        |   |                |            |      |     |                  |     |        |    |          |               |         |            |
|--------|---|----------------|------------|------|-----|------------------|-----|--------|----|----------|---------------|---------|------------|
| Menu   | 0 | NORMALIZATION: | per_kernel | GCD: | ALL | DISPATCH FILTER: | ALL | TOP N: | 10 | KERNELS: | matmul_kernel | 152.00  | Report Bug |
| 52.00  |   |                |            |      |     |                  |     | 52.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 53.00  |   |                |            |      |     |                  |     | 53.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 54.00  |   |                |            |      |     |                  |     | 54.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 55.00  |   |                |            |      |     |                  |     | 55.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 56.00  |   |                |            |      |     |                  |     | 56.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 57.00  |   |                |            |      |     |                  |     | 57.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 58.00  |   |                |            |      |     |                  |     | 58.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 59.00  |   |                |            |      |     |                  |     | 59.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 60.00  |   |                |            |      |     |                  |     | 60.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 61.00  |   |                |            |      |     |                  |     | 61.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 62.00  |   |                |            |      |     |                  |     | 62.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 63.00  |   |                |            |      |     |                  |     | 63.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 64.00  |   |                |            |      |     |                  |     | 64.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 65.00  |   |                |            |      |     |                  |     | 65.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 66.00  |   |                |            |      |     |                  |     | 66.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 67.00  |   |                |            |      |     |                  |     | 67.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 68.00  |   |                |            |      |     |                  |     | 68.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 69.00  |   |                |            |      |     |                  |     | 69.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 70.00  |   |                |            |      |     |                  |     | 70.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 71.00  |   |                |            |      |     |                  |     | 71.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 72.00  |   |                |            |      |     |                  |     | 72.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 73.00  |   |                |            |      |     |                  |     | 73.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 74.00  |   |                |            |      |     |                  |     | 74.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 75.00  |   |                |            |      |     |                  |     | 75.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 76.00  |   |                |            |      |     |                  |     | 76.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 77.00  |   |                |            |      |     |                  |     | 77.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 78.00  |   |                |            |      |     |                  |     | 78.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 79.00  |   |                |            |      |     |                  |     | 79.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 80.00  |   |                |            |      |     |                  |     | 80.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 81.00  |   |                |            |      |     |                  |     | 81.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 82.00  |   |                |            |      |     |                  |     | 82.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 83.00  |   |                |            |      |     |                  |     | 83.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 84.00  |   |                |            |      |     |                  |     | 84.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 85.00  |   |                |            |      |     |                  |     | 85.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 86.00  |   |                |            |      |     |                  |     | 86.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 87.00  |   |                |            |      |     |                  |     | 87.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 88.00  |   |                |            |      |     |                  |     | 88.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 89.00  |   |                |            |      |     |                  |     | 89.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 90.00  |   |                |            |      |     |                  |     | 90.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 91.00  |   |                |            |      |     |                  |     | 91.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 92.00  |   |                |            |      |     |                  |     | 92.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 93.00  |   |                |            |      |     |                  |     | 93.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 94.00  |   |                |            |      |     |                  |     | 94.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 95.00  |   |                |            |      |     |                  |     | 95.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 96.00  |   |                |            |      |     |                  |     | 96.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 97.00  |   |                |            |      |     |                  |     | 97.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 98.00  |   |                |            |      |     |                  |     | 98.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 99.00  |   |                |            |      |     |                  |     | 99.00  |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 100.00 |   |                |            |      |     |                  |     | 100.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 101.00 |   |                |            |      |     |                  |     | 101.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 102.00 |   |                |            |      |     |                  |     | 102.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 103.00 |   |                |            |      |     |                  |     | 103.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 104.00 |   |                |            |      |     |                  |     | 104.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 105.00 |   |                |            |      |     |                  |     | 105.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 106.00 |   |                |            |      |     |                  |     | 106.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 107.00 |   |                |            |      |     |                  |     | 107.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 108.00 |   |                |            |      |     |                  |     | 108.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 109.00 |   |                |            |      |     |                  |     | 109.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 110.00 |   |                |            |      |     |                  |     | 110.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 111.00 |   |                |            |      |     |                  |     | 111.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 112.00 |   |                |            |      |     |                  |     | 112.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 113.00 |   |                |            |      |     |                  |     | 113.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 114.00 |   |                |            |      |     |                  |     | 114.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 115.00 |   |                |            |      |     |                  |     | 115.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 116.00 |   |                |            |      |     |                  |     | 116.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 117.00 |   |                |            |      |     |                  |     | 117.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 118.00 |   |                |            |      |     |                  |     | 118.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 119.00 |   |                |            |      |     |                  |     | 119.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 120.00 |   |                |            |      |     |                  |     | 120.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 121.00 |   |                |            |      |     |                  |     | 121.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 122.00 |   |                |            |      |     |                  |     | 122.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 123.00 |   |                |            |      |     |                  |     | 123.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 124.00 |   |                |            |      |     |                  |     | 124.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 125.00 |   |                |            |      |     |                  |     | 125.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 126.00 |   |                |            |      |     |                  |     | 126.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |
| 127.00 |   |                |            |      |     |                  |     | 127.00 |    | 1152.00  | 1152.00       | 1152.00 | 1152.00    |